Signorini's Problem

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Contents

Abstract

This is the abstract.[1]

1 Introduction

This is the Introduction.

2 Mathematical Framework

This section introduces the basic notation used throughout the mathematical framework and the later analysis of Signorini's problem. Our goal is to fix a consistent language for bulk quantities, boundary quantities, and tensor-valued fields before introducing weak derivatives and Sobolev spaces. Unless stated otherwise, let $\Omega \subset \mathbb{R}^d$, $d \geq 1$, be a nonempty open set. Whenever trace operators or boundary spaces are used, we additionally assume that Ω is a bounded Lipschitz domain (see e.g. [1, Section 2.3]) and write $\Gamma := \partial\Omega$.

2.1 Scalars, Vectors and Matrices

The purpose of this subsection is mainly to introduce the notation used throughout this report. For $m, n \geq 1$, we write $\mathbb{R}^{m\times n}$ for the space of real $m \times n$ matrices and for vectors $u, v \in \mathbb{R}^m$, we denote by

$$u \cdot v := \sum_{j=1}^{m} u_j\, v_j, \qquad |u| := \sqrt{u \cdot u}$$

the Euclidean inner product and norm. For matrices $A, B \in \mathbb{R}^{m\times n}$, we use the Frobenius product

$$A : B := \sum_{j=1}^{m}\sum_{k=1}^{n} A_{jk}B_{jk}, \qquad |A| := \sqrt{A : A}.$$

The space $\mathrm{Sym}^2(\mathbb{R}^d)$ of symmetric second order tensors is

$$\mathrm{Sym}^2(\mathbb{R}^d) := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\}.$$

We will interpret these tensors as symmetric matrices in $\mathbb{R}^{d\times d}$ in the canonical basis of $\mathbb{R}^d$. Indeed,

$$\mathrm{Sym}^2(\mathbb{R}^d) \cong \left\{ S \in \mathbb{R}^{d\times d} \;\middle|\; S^\top = S \right\}.$$

Furthermore, if $\mathcal{A} : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$ is a fourth order tensor and $s \in \mathrm{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d}\sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{kl}, \qquad s : \mathcal{A}(x) : s := \sum_{i=1}^{d}\sum_{j=1}^{d}\sum_{k=1}^{d}\sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

In the remaining subsections of the mathematical framework section, we will denote by X the spaces $\mathbb{R}^m$ with $m \geq 1$ and talk about functions $f : \Omega \to X$. By the identification $\mathbb{R}^{m\times n} \cong \mathbb{R}^{mn}$ this also covers the case of functions with values in $\mathrm{Sym}^2(\mathbb{R}^d) \subset \mathbb{R}^{d\times d}$, as used later on in the main section. This identification is also isometric when considering Euclidean and Frobenius norm. If $X = \mathbb{R}$, i.e. $m = 1$, a space $\mathcal{X}(\Omega; X)$ of functions $f : \Omega \to X$ is always abbreviated by $\mathcal{X}(\Omega)$.

2.2 L^p-Spaces

We assume that the reader is familiar with the basic concept of L^p-spaces, but will nevertheless provide a brief introduction to establish the notations used here. We omit the case $p = \infty$, since it is not needed here. L^p-spaces on Ω are spaces of equivalence classes of p-integrable functions whose difference vanishes almost everywhere on Ω. Although this is the most correct way to define L^p-spaces, we will subsequently always assume an actual function by choosing an arbitrary representative of a class. The assumed measure on Ω here is the Lebesgue measure in $\mathbb{R}^d$. We record this in the following definition.

Definition 2.1 (L^p-Spaces). *Let $1 \le p < \infty$. The space of* `p-integrable` *functions on Ω is defined as*

$$L^p(\Omega; X) := \left\{ f : \Omega \to \mathbb{R}^m \text{ measurable } \middle| \int_\Omega |f|^p \, \mathrm{d}x < \infty \right\} \Big/ \sim,$$

where $f \sim g$ if and only if $f0g$ almost everywhere (short a.e.) in Ω. For some equivalence class $[f] \in L^p(\Omega; X)$ we choose an arbitrary representative and write $f = [f]$. Moreover, we equip $L^p(\Omega; X)$ with the norm

$$\|f\|_{0,p,\Omega} := \left(\int_\Omega |f|^p \, \mathrm{d}x \right)^{1/p}, \qquad u \in L^p(\Omega; X). \tag{2.1}$$

The notation $\|\cdot\|_{0,p,\Omega}$ will be more clear after Sobolev spaces are introduced as $L^p(\Omega; X)$ is exactly the Sobolev space of zero order. Of course, the space $L^p(\Gamma_0; X)$ of p-integrable functions on measurable and relatively open boundary parts $\Gamma_0 \subset \Gamma$ is defined analogously with the difference, that here the surface measure is used and the integral (2.1) is taken over Γ_0, i.e.

$$\|g\|_{0,p,\Gamma_0} := \left(\int_{\Gamma_0} |g|^p \, \mathrm{d}s \right)^{1/p}, \qquad g \in L^p(\Gamma_0; X).$$

The particular case $p = 2$ is of special interest, since it forms a Hilbert space.

Proposition 2.1 (L^2 is a Hilbert space). *The space $L^2(\Omega; X)$ together with the inner product*

$$(f,\, g)_{0,\Omega} := \int_\Omega f \cdot g \, \mathrm{d}x, \qquad f, g \in L^2(\Omega; X)$$

is a Hilbert space and the induced norm is

$$\|f\|_{0,\Omega} := \|f\|_{0,2,\Omega} = \left(\int_\Omega |f|^2 \, \mathrm{d}x \right)^{1/2}, \qquad f \in L^2(\Omega; X).$$

The same holds for $L^2(\Gamma_0; X)$ respectively.

For later purpose, we also need to consider the subspace of locally integrable functions on Ω. Again, on Γ_0 the same definition can be applied with respect to the surface measure.

Definition 2.2 (Locally Integrable Functions). *The space of* `locally integrable` *functions on Ω is defined as*

$$L^1_{\mathrm{loc}}(\Omega; X) := \left\{ f : \Omega \to X \text{ measurable } \middle| f|_K \in L^1(K; X) \;\forall K \subset \Omega,\ K \text{ compact} \right\}.$$

2.3 Test Functions & Distributional Derivatives

Before introducing Sobolev spaces, we first formalize a weak notion of differentiation. The reason is that the functions arising in variational formulations of boundary value problems are typically not smooth enough to admit classical derivatives in the pointwise sense. Nevertheless, one still wants to differentiate such functions in a meaningful way, at least under an integral sign. The key observation is that in integration by parts, derivatives can be shifted from an unknown function to a smooth test function with compact support. This suggests defining derivatives indirectly through their action on test functions. The resulting objects are distributions, and distributional derivatives extend classical derivatives whenever the latter exist. We will restrict ourself here for the domain Ω but again, everything can be formulated for boundary parts $\Gamma_0 \subset \Gamma$ as well. For a sufficiently regular function $f = (f_1, \dots, f_m)^\top : \Omega \to X$ and a multi-index $\alpha \in \mathbb{N}_0^d$, we denote by

$$\mathrm{D}^\alpha f_j := \frac{\partial^{|\alpha|} f_j}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}} : \Omega \to \mathbb{R}, \qquad 1 \leq j \leq m$$

the partial derivatives of the j-th component $f_j : \Omega \to \mathbb{R}$ with respect to alfa. The gradient of f is the Jacobian-matrix

$$\nabla f := \left(\frac{\partial f_j}{\partial x_k} \right)_{1 \leq j \leq m,\, 1 \leq k \leq d} : \Omega \to \mathbb{R}^{m \times d}.$$

For $m = d$, the divergence of $f : \Omega \to X$ is

$$\operatorname{div} f := \sum_{j=1}^{d} \frac{\partial f_j}{\partial x_j} : \Omega \to \mathbb{R}.$$

Unfortunately this does not cover the case of matrix-valued fields $A : \Omega \to \mathbb{R}^{m \times d}$. Thus, we separately define

$$\operatorname{div} A := \left(\sum_{k=1}^{d} \frac{\partial A_{jk}}{\partial x_k} \right)_{1 \leq j \leq m} : \Omega \to \mathbb{R}^m.$$

We collect continuous functions with continuous partial derivatives of the same order in the following well-known function spaces.

Definition 2.3 (Spaces of Continuously Differentiable Functions)**.** *We denote by $\mathcal{C}^0(\Omega)$ the space of continuous functions $u : \Omega \to \mathbb{R}$ and define $\mathcal{C}^0(\Omega; X) := (\mathcal{C}^0(\Omega))^m$ as the space of continuous functions $f : \Omega \to \mathbb{R}^m$. The space of* `continuously differentiable` *functions $f : \Omega \to X$ up to order $0 \leq k < \infty$ then is*

$$\mathcal{C}^k(\Omega; X) := \left\{ f \in \mathcal{C}^0(\Omega; X) \;\middle|\; |\alpha| \leq k \;\Rightarrow\; \mathrm{D}^\alpha (f)_j \in \mathcal{C}^0(\Omega) \;\forall\, 1 \leq j \leq m \right\},$$

The space of `infinitely continuously differentiable` *functions $f : \Omega \to X$ is defined as*

$$\mathcal{C}^\infty(\Omega; X) := \bigcap_{k \in \mathbb{N}_0} \mathcal{C}^k(\Omega; X).$$

Again, the above definitions may be extended to functions on $\Gamma_0 \subset \Gamma$. A particular subspace of $\mathcal{C}^\infty(\Omega; X)$ is of special interest to us and the concept of distributional derivatives. This is the space of infinitely often continuously differentiable functions with compact support, also called *test functions.*

Definition 2.4 (Test Functions)**.** *The space $\mathcal{D}(\Omega; X)$ contains all infinitely often continuously differentiable functions with compact support in Ω, i.e.*

$$\mathcal{D}(\Omega; X) = \{\varphi \in \mathcal{C}^\infty(\Omega; X) \mid \operatorname{supp}(\varphi) \subset \Omega \;\text{ compact}\}$$

A function $\mathcal{D}(\Omega; X)$ is called a `test function`*.*

The space of test functions allows us to encode differentiation through integration by parts. This leads to distributions, which extend classical derivatives to nonsmooth functions and, more generally, to generalized functions.

Definition 2.5 (Distributions)**.** *The space of scalar-valued* `distributions` *on Ω is the topological dual space*

$$\mathcal{D}'(\Omega) := (\mathcal{D}(\Omega))'.$$

Thus, a scalar-valued distribution is a linear continuous functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$. For $T \in \mathcal{D}'(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, we denote the duality pairing by $\langle T, \varphi \rangle$. Furthermore, we define the space of vector-valued distributions componentwise by

$$\mathcal{D}'(\Omega; X) := (\mathcal{D}'(\Omega))^m .$$

For $T = (T_1, \dots, T_m) \in \mathcal{D}'(\Omega; X)$ and $\varphi = (\varphi_1, \dots, \varphi_m) \in \mathcal{D}(\Omega; X)$, we use the duality pairing

$$\langle T, \varphi \rangle := \sum_{j=1}^{m} \langle T_j, \varphi_j \rangle.$$

Finally, we can define derivatives on such functionals.

Definition 2.6 (Distributional Derivative)**.** *Let $T \in \mathcal{D}'(\Omega)$ and $\alpha \in \mathbb{N}_0^d$. The* `distributional derivative` *$\mathrm{D}^\alpha T \in \mathcal{D}'(\Omega)$ is defined by*

$$\langle \mathrm{D}^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, \mathrm{D}^\alpha \varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

Distributional derivatives of vector-valued and matrix-valued distributions are defined componentwise. In particular, for $T = (T_1, \dots, T_m) \in \mathcal{D}'(\Omega; X)$, we define

$$\mathrm{D}^\alpha T := (\mathrm{D}^\alpha T_1, \dots, \mathrm{D}^\alpha T_m) \in \mathcal{D}'(\Omega; X).$$

Using the componentwise definition of distributional derivatives, we can now extend the classical differential operators introduced above to distributions. For $1 \le k \le d$, let $e_k \in \mathbb{N}_0^d$ denote the k-th unit multi-index, i.e.

$$e_k := (0, \dots, 0, 1, 0, \dots, 0)^\top,$$

where the entry 1 is in the k-th position. Then D^{e_k} in the distributional sense corresponds to the first-order derivative in the x_k-direction. We thus write $\partial_{x_k} = \mathrm{D}^{e_k}$.

Definition 2.7 (Distributional Gradient and Divergence)**.** *Let $T = (T_1, \dots T_m)^\top \in \mathcal{D}'(\Omega; X)$ be a distribution.*

i) The `distributional gradient` *of T is*

$$\nabla T := \left(\partial_{x_j} T_k\right)^\top_{1 \le j \le d,\, 1 \le k \le m} \in \mathcal{D}'(\Omega; \mathbb{R}^{m \times d}).$$

ii) For $d = m$, the `distributional divergence` *of $T = (U_1, \dots, U_d)^\top$ is*

$$\operatorname{div} T := \sum_{k=1}^{d} \partial_{x_k} T_k \in \mathcal{D}'(\Omega).$$

At this stage, the above definitions are purely distributional and do not require any pointwise differentiability. In the next subsection, we will identify locally integrable functions with regular distributions and show that weak derivatives are precisely those distributional derivatives that are again induced by functions.

2.4 Regular Distributions & Weak Derivatives

The distributional framework introduced above becomes particularly useful once distributions are induced by locally integrable functions. These so-called regular distributions provide the link between generalized differentiation and the weak derivatives used later in Sobolev spaces and variational formulations.

Definition 2.8 (Regular and Singular Distributions). *A* `regular distribution` *is a distribution $T_f \in \mathcal{D}'(\Omega; X)$ induced by a locally integrable function $f \in L^1_{\mathrm{loc}}(\Omega; X)$ by*

$$\langle T_f, \varphi \rangle := \int_\Omega f \cdot \varphi \, \mathrm{d}x, \qquad \forall \varphi \in \mathcal{D}(\Omega; X).$$

In particular, $L^1_{\mathrm{loc}}(\Omega; X)$ is embedded into $\mathcal{D}'(\Omega; X)$ via $f \mapsto T_f$ and its image is the space of regular distributions. Distributions that are not regular are called `singular distributions`*.*

The concept of distributional derivatives applied to regular distributions is exactly the concept of weak derivatives.

Definition 2.9 (Weak Derivative). *Let $f \in L^1_{\mathrm{loc}}(\Omega; X)$ and $\alpha \in \mathbb{N}_0^d$. A locally integrable function $g \in L^1_{\mathrm{loc}}(\Omega; X)$ is called the* `weak derivative` *of f of order α if*

$$\int_\Omega g \cdot \varphi \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega f \cdot \mathrm{D}^\alpha \varphi \, \mathrm{d}x \qquad \forall \varphi \in \mathcal{D}(\Omega; X),$$

where D^α is understood in the classical sense. In this case, we write $g = \mathrm{D}^\alpha f$.

Proposition 2.2. *Let $f, g \in L^1_{\mathrm{loc}}(\Omega; X)$. Then g is the weak derivative $\mathrm{D}^\alpha f$ of f if and only if*

$$\langle \mathrm{D}^\alpha T_f, \varphi \rangle = \langle T_g, \varphi \rangle \qquad \forall \varphi \in \mathcal{D}(\Omega; X),$$

or in other words $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$.

Proof. Assume first that $g = \mathrm{D}^\alpha f$ in the weak sense. Then for every $\varphi \in \mathcal{D}(\Omega; X)$, by definition of the distributional derivative and the weak derivative,

$$\langle \mathrm{D}^\alpha T_f, \varphi \rangle = (-1)^{|\alpha|} \langle T_f, \mathrm{D}^\alpha \varphi \rangle = (-1)^{|\alpha|} \int_\Omega (f, \mathrm{D}^\alpha \varphi)_X \, \mathrm{d}x = \int_\Omega (g, \varphi)_X \, \mathrm{d}x = \langle T_g, \varphi \rangle.$$

Hence $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$. Conversely, assume that $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$. Then for every test function $\varphi \in \mathcal{D}(\Omega; X)$,

$$\int_\Omega (g, \varphi)_X \, \mathrm{d}x = \langle T_g, \varphi \rangle = \langle \mathrm{D}^\alpha T_f, \varphi \rangle = (-1)^{|\alpha|} \langle T_f, \mathrm{D}^\alpha \varphi \rangle = (-1)^{|\alpha|} \int_\Omega (f, \mathrm{D}^\alpha \varphi)_X \, \mathrm{d}x.$$

Thus, g is the weak derivative $\mathrm{D}^\alpha f$ of f. ■

Finally, we want to put the distributional derivatives of regular distributions in relation to the distributional derivatives from the last section.

Proposition 2.3 (Integration by Parts for Regular Distributions).

i) Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let $T_u \in \mathcal{D}'(\Omega)$ be the induced regular distribution. Then for every $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$,

$$\langle \nabla T_u, \varphi \rangle = -\langle T_u, \operatorname{div} \varphi \rangle = -\int_\Omega u \operatorname{div} \varphi \, \mathrm{d}x,$$

where $\operatorname{div} \varphi$ is understood in the classical sense.

ii) Let $F \in L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^d)$ and let $T_F \in \mathcal{D}'(\Omega; \mathbb{R}^d)$ be the induced regular distribution. Then *for every* $\varphi \in \mathcal{D}(\Omega)$,

$$\langle \operatorname{div} T_F,\, \varphi\rangle = -\langle T_F,\, \nabla\varphi\rangle = -\int_\Omega F\cdot\nabla\varphi\,\mathrm{d}x,$$

where $\nabla\varphi$ *is understood in the classical sense.*

iii) Let $A \in L^1_{\mathrm{loc}}(\Omega;\mathbb{R}^{m\times d})$ *and let* $T_A \in \mathcal{D}'(\Omega;\mathbb{R}^{m\times d})$ *be the induced regular distribution. Then for every* $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^m)$,

$$\langle \operatorname{div} T_A,\, \varphi\rangle = -\int_\Omega A : \nabla\varphi\,\mathrm{d}x,$$

where $\nabla\varphi = (\partial_{x_k}\varphi_j)_{1\le j\le m}^{1\le k\le d}$ *is the Jacobian in the classical sense.*

Proof. i) Let $\varphi = (\varphi_1,\dots,\varphi_d)^\top \in \mathcal{D}(\Omega;\mathbb{R}^d)$. By the componentwise definition of the pairing and of the distributional gradient,

$$\langle \nabla T_u,\, \varphi\rangle = \sum_{k=1}^d \langle \partial_{x_k} T_u,\, \varphi_k\rangle.$$

Using the definition of the distributional derivative, we obtain

$$\langle \nabla T_u,\, \varphi\rangle = -\sum_{k=1}^d \langle T_u,\, \partial_{x_k}\varphi_k\rangle = -\left\langle T_u,\, \sum_{k=1}^d \partial_{x_k}\varphi_k\right\rangle = -\langle T_u,\, \operatorname{div}\varphi\rangle.$$

Since T_u is induced by u, this is

$$\langle \nabla T_u,\, \varphi\rangle = -\int_\Omega u\,\operatorname{div}\varphi\,\mathrm{d}x.$$

ii) Let $\varphi \in \mathcal{D}(\Omega)$. By definition of the distributional divergence and the distributional derivative,

$$\langle \operatorname{div} T_F,\, \varphi\rangle = \sum_{k=1}^d \langle \partial_{x_k}(T_F)_k,\, \varphi\rangle = -\sum_{k=1}^d \langle (T_F)_k,\, \partial_{x_k}\varphi\rangle.$$

Using that $(T_F)_k$ is induced by F_k, we obtain

$$\langle \operatorname{div} T_F,\, \varphi\rangle = -\sum_{k=1}^d \int_\Omega F_k\,\partial_{x_k}\varphi\,\mathrm{d}x = -\int_\Omega F\cdot\nabla\varphi\,\mathrm{d}x.$$

This is the stated identity.

iii) Let $\varphi = (\varphi_1,\dots,\varphi_m)^\top \in \mathcal{D}(\Omega;\mathbb{R}^m)$. Using the row-wise definition of $\operatorname{div} T_A$ and the definition of distributional derivatives, we get

$$\langle \operatorname{div} T_A,\, \varphi\rangle = \sum_{j=1}^m \left\langle \sum_{k=1}^d \partial_{x_k}(T_A)_{jk},\, \varphi_j\right\rangle = -\sum_{j=1}^m\sum_{k=1}^d \langle (T_A)_{jk},\, \partial_{x_k}\varphi_j\rangle.$$

Since T_A is induced by A, this yields

$$\langle \operatorname{div} T_A,\, \varphi\rangle = -\sum_{j=1}^m\sum_{k=1}^d \int_\Omega A_{jk}\,\partial_{x_k}\varphi_j\,\mathrm{d}x = -\int_\Omega A:\nabla\varphi\,\mathrm{d}x.$$

■

2.5 Sobolev Spaces

In this subsection we restrict attention to the Hilbert case $p = 2$, since this is the only case needed for the subsequent variational analysis. All corresponding constructions can also be carried out for general $1 \le p < \infty$ using the spaces $W^{k,p}$, but we will not pursue this here. Before defining Sobolev spaces, we connect the previously introduced distributional framework with the L^2-spaces from Section 2.2. Since Sobolev spaces are defined by requiring distributional derivatives to belong again to $L^2(\Omega; X)$, it is essential that L^2-functions can be interpreted as regular distributions.

Lemma 2.1 (Cauchy–Schwarz Inequality)**.** *For all $f, g \in L^2(\Omega; X)$ is holds that*

$$\left| \int_\Omega f \cdot g \,\mathrm{d}x \right| \le \|f\|_{0,\Omega} \, \|g\|_{0,\Omega}.$$

$\square$

Proposition 2.4 (L^2-Functions are Locally Integrable)**.** *It holds that $L^2(\Omega; X) \subset L^1_{\mathrm{loc}}(\Omega; X)$. In particular, every $f \in L^2(\Omega; X)$ induces a regular distribution $T_f \in \mathcal{D}'(\Omega; X)$ and admits distributional derivatives of arbitrary order.*

Proof. Let $f \in L^2(\Omega; X)$ and let $K \subset \Omega$ be compact. Since K has finite Lebesgue measure, the Cauchy–Schwarz inequality yields

$$\int_K |f| \,\mathrm{d}x \le \left(\int_K |f|^2 \,\mathrm{d}x \right)^{1/2} \left(\int_K 1^2 \,\mathrm{d}x \right)^{1/2} = \|f\|_{0,K} \, |K|^{1/2} \le \|f\|_{0,\Omega} \, |K|^{1/2} < \infty.$$

Hence $f|_K \in L^1(K; X)$. Since $K \subset \Omega$ was arbitrary compact, we conclude that $f \in L^1_{\mathrm{loc}}(\Omega; X)$. The remaining statement follows from the embedding $L^1_{\mathrm{loc}}(\Omega; X) \hookrightarrow \mathcal{D}'(\Omega; X)$ via $f \mapsto T_f$ and the fact that $\mathrm{D}^\alpha T_f$ is defined for every $\alpha \in \mathbb{N}_0^d$. $\blacksquare$

By Proposition 2.4, every $f \in L^2(\Omega; X)$ can be identified with the regular distribution $T_f \in \mathcal{D}'(\Omega; X)$ given by

$$\langle T_f, \varphi \rangle = \int_\Omega f \cdot \varphi \,\mathrm{d}x, \qquad \forall \, \varphi \in \mathcal{D}(\Omega; X).$$

In the following, we will use this identification implicitly whenever no confusion can arise. Moreover, since $\varphi \in \mathcal{D}(\Omega; X)$ has compact support, we have $\varphi \in L^2(\Omega; X)$ and therefore $\langle T_f, \varphi \rangle = (f, \varphi)_{0,\Omega}$. Thus, on test functions, the duality pairing of the regular distribution T_f coincides with the L^2-inner product. In the Sobolev setting, we therefore use the terms distributional derivative and weak derivative interchangeably whenever the derivative is represented by an L^2-function.

Definition 2.10 (Sobolev Spaces of Integer Order)**.** *Let $k \in \mathbb{N}_0$. The* `Sobolev space` *$H^k(\Omega; X)$ is defined by*

$$H^k(\Omega; X) := \left\{ u \in L^2(\Omega; X) \;\middle|\; \mathrm{D}^\alpha u \in L^2(\Omega; X) \;\forall \, \alpha \in \mathbb{N}_0^d, \; |\alpha| \le k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. Moreover, we equip $H^k(\Omega; X)$ with the norm

$$\|u\|_{k,\Omega} := \left(\sum_{|\alpha| \le k} \|\mathrm{D}^\alpha u\|_{0,\Omega}^2 \right)^{1/2}, \qquad u \in H^k(\Omega; X).$$

We adopt the usual convention $H^0(\Omega; X) = L^2(\Omega; X)$ explaining the notion of the L^2-norm. Furthermore, $H^k(\Omega; X)$ is a Hilbert space.

Proposition 2.5 ($H^k(\Omega; X)$ is a Hilbert Space)**.** *The Sobolev space $H^k(\Omega; X)$ together with the inner product*

$$(u,\, v)_{k,\Omega} := \sum_{|\alpha|\le k} (\mathrm{D}^\alpha u,\, \mathrm{D}^\alpha v)_{0,\Omega}, \qquad u, v \in H^k(\Omega; X)$$

is a Hilbert space and the induced the norm $\|\cdot\|_{k,\Omega}$. □

After the integer-order spaces $H^k(\Omega; X)$, we also need Sobolev spaces of non-integer order. The main reason is that trace operators naturally lead to boundary spaces of fractional order. In particular, traces of $H^1(\Omega; X)$-functions are, in general, not measured in another integer-order Sobolev space, but in a space of order $1/2$ on the boundary and similarly on boundary parts $\Gamma_0 \subset \Gamma$. For a non-integer order $s = k + \theta$ with $\theta \in (0, 1)$, the remaining fractional part θ is measured by difference quotients. More precisely, one measures how strongly a function oscillates between pairs of points $x, y \in \Omega$, with a weight that becomes singular as $|x - y| \to 0$. This leads to the following quantity.

Definition 2.11 (Slobodeckij Seminorm)**.** *Let $m \ge 1$ and $\theta \in (0, 1)$. For a measurable function $u : \Omega \to X$, we define the* `Slobodeckij semi-norm`

$$|u|_{\theta,\Omega} := \left(\int_\Omega \int_\Omega \frac{|u(x) - u(y)|^2}{|x - y|^{d+2\theta}} \,\mathrm{d}x \,\mathrm{d}y \right)^{1/2},$$

whenever the right-hand side is finite.

The quantity $|u|_{\theta,\Omega}$ is only a semi-norm, since it vanishes on constant functions. Combined with an L^2-term, however, it yields a genuine norm.

Definition 2.12 (Sobolev Spaces of Fractional Order)**.** *Let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. The* `Sobolev space` *of fractional order s is defined by*

$$H^s(\Omega; X) := \left\{ u \in H^k(\Omega; X) \;\middle|\; |\mathrm{D}^\alpha u|_{\theta,\Omega} < \infty \;\forall\, \alpha \in \mathbb{N}_0^d,\; |\alpha| = k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. Moreover, we equip $H^s(\Omega; X)$ with the norm

$$\|u\|_{s,\Omega} := \left(\|u\|_{k,\Omega}^2 + |u|_{s,\Omega}^2 \right)^{1/2}, \qquad u \in H^s(\Omega; X).$$

Proposition 2.6 ($H^s(\Omega; X)$ is a Hilbert Space)**.** *Let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. Then $H^s(\Omega; X)$ is a Hilbert space with inner product*

$$(u,\, v)_{s,\Omega} := (u,\, v)_{k,\Omega} + \sum_{|\alpha|=k} \int_\Omega \int_\Omega \frac{(\mathrm{D}^\alpha u(x) - \mathrm{D}^\alpha u(y)) \cdot (\mathrm{D}^\alpha v(x) - \mathrm{D}^\alpha v(y))}{|x - y|^{d+2\theta}} \,\mathrm{d}x \,\mathrm{d}y$$

for $u, v \in H^s(\Omega; X)$. The induced norm is $\|\cdot\|_{s,\Omega}$. □

2.6 Trace Operators and Boundary Spaces

In this subsection, we assume that $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain and write $\Gamma := \partial\Omega$. We start with the following definition.

Definition 2.13 (Classical Trace on Smooth Functions)**.** *We define the* `classical trace operator`

$$\gamma_0 : \mathcal{C}^\infty(\overline{\Omega}; X) \to L^2(\Gamma; X), \qquad \gamma_0 u := u|_\Gamma$$

on the space

$$\mathcal{C}^\infty(\overline{\Omega}; X) := \left\{ u|_{\overline{\Omega}} \;\middle|\; u \in \mathcal{C}^\infty(U; X) \text{ for some open neighborhood } U \supset \overline{\Omega} \right\}.$$

In variational formulations, boundary conditions are imposed on traces of bulk functions. For $u \in H^s(\Omega; X)$, pointwise boundary values are in general not defined. The trace theorem provides a canonical replacement.

Definition 2.14 (Boundary Sobolev Spaces)**.** *Let $s > 1/2$. We define*

$$H^{s-1/2}(\Gamma; X) := \gamma_0(\mathcal{C}^\infty(\overline{\Omega}; X))$$

equipped with the norm

$$\|g\|_{s-\frac{1}{2},\Gamma} := \inf \left\{ \|u\|_{s,\Omega} \;\middle|\; u \in \mathcal{C}^\infty(\overline{\Omega}; X),\ \gamma_0 u = g \right\}.$$

The underlying set is the same for different values of $s > 1/2$, but it is equipped with different norms.

Theorem 2.1 (Trace Theorem)**.** *Let $1/2 < s \leq 1$. Then the classical trace operator γ_0 extends uniquely to a continuous surjective linear operator*

$$\gamma : H^s(\Omega; X) \to H^{s-1/2}(\Gamma; X).$$

In particular, there exists a constant $C_{\mathrm{tr},s} > 0$ such that

$$\|\gamma u\|_{s-\frac{1}{2},\Gamma} \leq C_{\mathrm{tr},s} \|u\|_{s,\Omega} \qquad \forall\, u \in H^s(\Omega; X).$$

In the special case $s = 1$, one has

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; X)}^{\|\cdot\|_{1,\Omega}} =: H_0^1(\Omega; X).$$

In the particularly important case $s = 1$, the trace space $H^{1/2}(\Gamma; X)$ is precisely the set of boundary values attained by $H^1(\Omega; X)$-functions. Since $H^{1/2}(\Gamma; X) \subset L^2(\Gamma; X)$, the trace γu may be viewed as an L^2-boundary function, while its natural norm is the quotient norm induced by $H^1(\Omega; X)$. We want to formulate similar definitions on boundary parts $\Gamma_0 \subset \Gamma$ and have a notion for their topological dual spaces.

Definition 2.15 (Boundary Spaces on Boundary Parts)**.** *Let $\Gamma_0 \subset \Gamma$ be relatively open and let $s > 1/2$. We define*

$$H^{s-1/2}(\Gamma_0; X) := \left\{ g : \Gamma_0 \to X \;\middle|\; \exists\, \widetilde{g} \in H^{s-1/2}(\Gamma; X) \text{ with } \widetilde{g}|_{\Gamma_0} = g \right\},$$

with norm

$$\|g\|_{s-\frac{1}{2},\Gamma_0} := \inf\left\{\|\tilde{g}\|_{s-\frac{1}{2},\Gamma} \,\middle|\, \tilde{g} \in H^{s-1/2}(\Gamma;X),\ \tilde{g}|_{\Gamma_0} = g\right\}.$$

Moreover, we define

$$H_{00}^{s-1/2}(\Gamma_0;X) := \left\{g \in H^{s-1/2}(\Gamma_0;X) \,\middle|\, \mathrm{Ext}_0(g) \in H^{s-1/2}(\Gamma;X)\right\},$$

where $\mathrm{Ext}_0(g)$ *denotes the zero extension of* g *from* Γ_0 *to* Γ*, i.e.*

$$\mathrm{Ext}_0(g)(x) = \begin{cases} g(x), & x \in \Gamma_0, \\ 0, & x \in \Gamma \setminus \Gamma_0. \end{cases}$$

We equip $H_{00}^{s-1/2}(\Gamma_0;X)$ *with the norm*

$$\|g\|_{s-\frac{1}{2},00,\Gamma_0} := \|\mathrm{Ext}_0(g)\|_{s-\frac{1}{2},\Gamma}.$$

Definition 2.16 (Dual Spaces on Γ and on Γ_0). *We define*

$$H^{-1/2}(\Gamma;X) := (H^{1/2}(\Gamma;X))'$$

and

$$H^{-1/2}(\Gamma_0;X) := (H_{00}^{1/2}(\Gamma_0;X))'.$$

The corresponding duality pairings are denoted by $\langle\cdot,\cdot\rangle_\Gamma$ *and* $\langle\cdot,\cdot\rangle_{\Gamma_0}$*, respectively.*

Theorem 2.2 (Lifting Operator). *Let* $\Omega \subset \mathbb{R}^d$ *be a bounded Lipschitz domain and let* $\Gamma_0 \subset \Gamma$ *be relatively open.*

i) There exists a bounded linear operator

$$\mathcal{L} : H^{1/2}(\Gamma;\mathbb{R}^d) \to H^1(\Omega;\mathbb{R}^d)$$

such that

$$\gamma(\mathcal{L}g) = g \qquad \forall\, g \in H^{1/2}(\Gamma;\mathbb{R}^d),$$

and

$$\|\mathcal{L}g\|_{1,\Omega} \le C\|g\|_{1/2,\Gamma}.$$

ii) There exists a bounded linear operator

$$\mathcal{L} : H_{00}^{1/2}(\Gamma_0;\mathbb{R}^d) \to H^1(\Omega;\mathbb{R}^d)$$

such that for every $g \in H_{00}^{1/2}(\Gamma_0;\mathbb{R}^d)$,

$$\gamma(\mathcal{L}g)|_{\Gamma_0} = g \qquad \textit{and} \qquad \gamma(\mathcal{L}g) = 0 \textit{ on } \Gamma \setminus \Gamma_0,$$

and

$$\|\mathcal{L}g\|_{1,\Omega} \le C\|g\|_{H_{00}^{1/2}(\Gamma_0)}.$$

2.7 $H(\mathrm{div};\Omega)$ and Normal Traces

In elasticity, stress fields are (typically symmetric) matrix-valued functions. While the assumption $\tau \in L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d))$ is sufficient to interpret volume integrals of the form

$$\int_\Omega \tau : \nabla v \,\mathrm{d}x,$$

it does not provide a meaningful boundary traction $\tau\,\mathrm{n}$ on Γ in general. The reason is that for $\tau \in L^2(\Omega;\mathbb{R}^{d\times d})$ no pointwise boundary trace is available, so the product $\tau\,\mathrm{n}$ cannot be defined as an $L^2(\Gamma;\mathbb{R}^d)$-function in general. In contrast, for $v \in H^1(\Omega;\mathbb{R}^d)$ the trace γv is well-defined by the trace theorem, and hence the normal component

$$v_\mathrm{n} := (\gamma v)\cdot \mathrm{n}$$

makes sense as an element of $L^2(\Gamma)$ (and later on boundary parts such as $\Gamma_C \subset \Gamma$). For stresses, however, one has to proceed differently and define boundary tractions in a weak sense via integration by parts. A natural additional regularity assumption is $\tau \in H(\mathrm{div};\Omega)$, which ensures that $\operatorname{div}\tau$ exists as an L^2-function.

Definition 2.17 ($H(\mathrm{div};\Omega)$)**.** *We define*

$$H(\mathrm{div};\Omega) := \left\{\tau \in L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d)) \;\middle|\; \operatorname{div}\tau \in L^2(\Omega;\mathbb{R}^d)\right\},$$

where $\operatorname{div}\tau$ *is understood row-wise in the weak sense.*

For smooth fields one has the classical Green identity. Here $\mathrm{n} : \Gamma \to \mathbb{R}^d$ denotes the outward unit normal, which exists ds-almost everywhere on Lipschitz boundaries.

Proposition 2.7 (Integration by Parts)**.** *If* $\tau \in C^1(\overline{\Omega};\mathbb{R}^{d\times d})$ *and* $v \in C^1(\overline{\Omega};\mathbb{R}^d)$*, then*

$$\int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x = \int_\Gamma (\tau\,\mathrm{n})\cdot v \,\mathrm{d}s. \tag{2.2}$$

The key point is that the left-hand side of (2.2) still makes sense for $\tau \in H(\mathrm{div};\Omega)$ and $v \in H^1(\Omega;\mathbb{R}^d)$, by interpreting $\operatorname{div}\tau$ and ∇v in the weak sense. Hence, for fixed $\tau \in H(\mathrm{div};\Omega)$ we define

$$F_\tau(v) := \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x, \qquad v \in H^1(\Omega;\mathbb{R}^d). \tag{2.3}$$

By Cauchy–Schwarz,

$$|F_\tau(v)| \le (\|\tau\|_{0,\Omega} + \|\operatorname{div}\tau\|_{0,\Omega})\,\|v\|_{1,\Omega} \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d),$$

i.e. F_τ is a continuous linear functional on $H^1(\Omega;\mathbb{R}^d)$.

Proposition 2.8 (Dependence of F_τ on the Trace)**.** *Let* $\tau \in H(\mathrm{div};\Omega)$*. Then* F_τ *vanishes on* $H^1_0(\Omega;\mathbb{R}^d)$*, i.e.*

$$F_\tau(w) = 0 \qquad \forall\, w \in H^1_0(\Omega;\mathbb{R}^d).$$

Consequently, F_τ *depends only on the trace* γv *in the sense that*

$$\gamma v_1 = \gamma v_2 \quad \Longrightarrow \quad F_\tau(v_1) = F_\tau(v_2) \qquad \forall\, v_1, v_2 \in H^1(\Omega;\mathbb{R}^d).$$

Proof. Let $w \in H^1_0(\Omega;\mathbb{R}^d)$. By density, there exists a sequence $(w_m)_{m\in\mathbb{N}} \subset \mathcal{D}(\Omega;\mathbb{R}^d)$ such that

$$w_m \to w \qquad \text{in } H^1(\Omega;\mathbb{R}^d).$$

Since $\tau \in H(\mathrm{div};\Omega)$, we have for every $m \in \mathbb{N}$ that

$$\int_\Omega (\operatorname{div}\tau)\cdot w_m \,\mathrm{d}x = -\int_\Omega \tau : \nabla w_m \,\mathrm{d}x,$$

and therefore

$$F_\tau(w_m) = 0 \qquad \forall\, m \in \mathbb{N}.$$

Passing to the limit $m \to \infty$ and using the continuity of F_τ on $H^1(\Omega;\mathbb{R}^d)$ yields

$$F_\tau(w) = \lim_{m\to\infty} F_\tau(w_m) = 0.$$

Hence F_τ vanishes on $H_0^1(\Omega;\mathbb{R}^d)$. Now let $v_1, v_2 \in H^1(\Omega;\mathbb{R}^d)$ satisfy $\gamma v_1 = \gamma v_2$. Then

$$\gamma(v_1 - v_2) = 0.$$

By the characterization $\ker(\gamma) = H_0^1(\Omega;\mathbb{R}^d)$, we obtain $v_1 - v_2 \in H_0^1(\Omega;\mathbb{R}^d)$. Therefore,

$$F_\tau(v_1) - F_\tau(v_2) = F_\tau(v_1 - v_2) = 0,$$

which proves the claim. ∎

By Proposition 2.8, the functional F_τ depends only on the trace γv. Hence the map

$$\gamma v \mapsto F_\tau(v)$$

defines a well-defined continuous linear functional on $H^{1/2}(\Gamma;\mathbb{R}^d)$. By Definition 2.16, there therefore exists a unique element of $H^{-1/2}(\Gamma;\mathbb{R}^d)$ representing this functional.

Definition 2.18 (Normal Trace / Boundary Traction). *Let $\tau \in H(\mathrm{div};\Omega)$. The* `normal trace` *(or boundary traction) of τ is the unique element*

$$\gamma_{\mathrm{n}}(\tau) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$$

such that

$$\langle \gamma_{\mathrm{n}}(\tau),\, \gamma v\rangle_\Gamma = \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d). \tag{2.4}$$

In the smooth case, (2.4) reduces to (2.2), hence $\gamma_{\mathrm{n}}(\tau)$ coincides with the classical traction $\tau\,\mathrm{n}$. We also define the equivalent concept on a boundary part $\Gamma_0 \subset \Gamma$. Using the zero extension from Definition 2.14, one obtains in the same way a well-defined functional on $H_{00}^{1/2}(\Gamma_0;\mathbb{R}^d)$.

Definition 2.19 (Normal Trace on a Boundary Part). *Let $\Gamma_0 \subset \Gamma$ be relatively open and let $\tau \in H(\mathrm{div};\Omega)$. The normal trace on Γ_0 is defined as the unique element*

$$\gamma_{\mathrm{n},\Gamma_0}(\tau) \in H^{-1/2}(\Gamma_0;\mathbb{R}^d)$$

given by

$$\langle \gamma_{\mathrm{n},\Gamma_0}(\tau),\, g\rangle_{\Gamma_0} := \langle \gamma_{\mathrm{n}}(\tau),\, \mathrm{Ext}_0(g)\rangle_\Gamma, \qquad g \in H_{00}^{1/2}(\Gamma_0;\mathbb{R}^d),$$

where $\mathrm{Ext}_0(g)$ denotes the zero extension of g from Γ_0 to Γ.

3 Signorini's Problem

3.1 Physical and Geometric Setup of the Problem

We consider an elastic body occupying a reference configuration $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$. Under the action of external forces and possible contact with a rigid foundation, the body deforms. In the *small–strain* framework, the deformation is described by the *displacement field* $u : \Omega \to \mathbb{R}^d$, and a material point $x \in \Omega$ is mapped to its deformed position

$$\varphi(x) := x + u(x), \qquad x \in \overline{\Omega}.$$

The aim of Signorini's problem is to determine an *equilibrium* displacement u under standard boundary conditions and *unilateral* contact constraints on the part of the boundary that may touch the rigid foundation. Throughout the whole report, $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ denotes an open, bounded and connected *Lipschitz* domain. The later means, that the boundary $\Gamma := \partial\Omega$ can locally be described as the graph of a Lipschitz continuous function. This ensures for example, that the outward normal vector $\mathrm{n} : \Gamma : \mathbb{R}^d$ is well–defined almost everywhere. The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \,\dot{\cup}\, \overline{\Gamma_N} \,\dot{\cup}\, \overline{\Gamma_C}$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 1). On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D,

$$\forall x \in \Gamma_D : \ x + u(x) = x \qquad \Longrightarrow \qquad (\gamma u)|_{\Gamma_D} = 0.$$

The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. We assume that Γ_C and Γ_D have positive $(d-1)$–Lebesgue measure. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\mathrm{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral contact* through *nonpenetration* and *complementarity*.

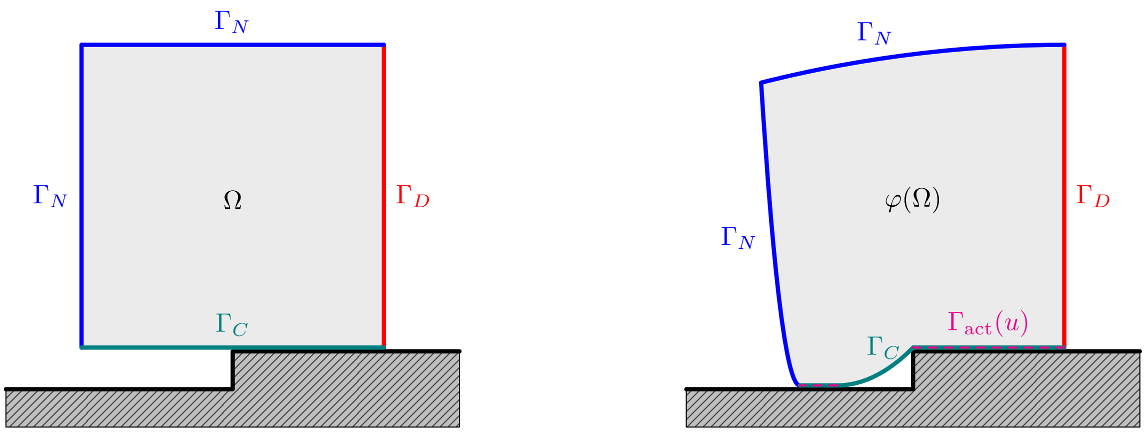

Figure 1: 2D example of Signorini's problem before (left) and after displacement (right).

3.2 Weak Formulation

3.2.1 Admissible Displacements

We start by introducing the energy space and the admissible set of displacements. Homogeneous Dirichlet boundary conditions on Γ_D are incorporated by

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \;\middle|\; \gamma v = 0 \text{ on } \Gamma_D \right\} \subset H^1(\Omega; \mathbb{R}^d). \tag{3.1}$$

For $v \in V$ we define the normal component of the boundary displacement on Γ_C by

$$v_{\mathrm{n}} := (\gamma v) \cdot \mathrm{n} \qquad \text{on } \Gamma_C, \tag{3.2}$$

where n is the outward unit normal on Γ. The unilateral nature of contact is expressed by the *nonpenetration* constraint. The rigid foundation is located on the exterior side of Γ_C in direction of n. Hence the body is not allowed to move across Γ_C into the obstacle. In the small–strain setting this yields that the solution must satisfy

$$u_{\mathrm{n}} \leq 0 \qquad \text{a.e. on } \Gamma_C.$$

This motivates the nonempty, closed and convex cone of admissible displacements

$$K := \{ v \in V \mid v_{\mathrm{n}} \leq 0 \text{ a.e. on } \Gamma_C \} \subset V. \tag{3.3}$$

Proposition 3.1 (K is a closed convex cone)**.** *The set K defined in 3.3 is a nonempty, closed and convex cone in V.*

Proof. K is nonempty, since $0 \in K$. Let $\lambda \in [0, \infty)$ and $v \in K \subset V$. Since V is a vector space, we have $\lambda v \in V$. Furthermore,

$$(\lambda v)_{\mathrm{n}} = \gamma(\lambda v) \cdot \mathrm{n} = (\lambda \gamma v) \cdot \mathrm{n} = \lambda(\gamma v \cdot n) = \lambda v_{\mathrm{n}}.$$

Since $v_{\mathrm{n}} \leq 0$ a.e. on Γ_C and $\lambda \geq 0$, we also have $\lambda v_{\mathrm{n}} \leq 0$ a.e. on Γ_C, i.e. $\lambda v \in K$. Thus K is a cone in V. Let $v, w \in K$ and $\theta \in [0, 1]$ and set $u := \theta v + (1 - \theta) w \in V$. Then we have

$$u_{\mathrm{n}} = (\gamma u) \cdot \mathrm{n} = (\theta \gamma v + (1 - \theta) \gamma w) \cdot \mathrm{n} = \theta(\gamma v \cdot \mathrm{n}) + (1 - \theta)(\gamma w \cdot \mathrm{n}) = \theta v_{\mathrm{n}} + (1 - \theta) w_{\mathrm{n}}$$

and $v_{\mathrm{n}}, w_{\mathrm{n}} \leq 0$ a.e. on Γ_C and $\theta, (1 - \theta) \geq 0$. Thus,

$$\theta v_{\mathrm{n}} + (1 - \theta) w_{\mathrm{n}} \leq 0 \qquad \text{a.e. on } \Gamma_C,$$

i.e. $u \in K$, which makes K convex. To prove that K is closed, let $(v_k)_{k \in \mathbb{N}} \subset K$ with $v_k \to v$ in V. Since the trace operator $\gamma : V \to L^2(\Gamma_C; \mathbb{R}^d)$ is continuous, the mapping

$$T : V \to L^2(\Gamma_C), \qquad T(w) := (\gamma w) \cdot \mathrm{n} = w_{\mathrm{n}},$$

is continuous as well. Hence,

$$(v_k)_{\mathrm{n}} = T(v_k) \to T(v) = v_{\mathrm{n}} \qquad \text{in } L^2(\Gamma_C).$$

Since $(v_k)_{\mathrm{n}} \leq 0$ a.e. on Γ_C for all k, there exists a subsequence $((v_{k_j})_{\mathrm{n}})_{j \in \mathbb{N}}$ such that

$$(v_{k_j})_{\mathrm{n}}(x) \to v_{\mathrm{n}}(x) \qquad \text{for a.e. } x \in \Gamma_C.$$

Passing to the limit in the inequality $(v_{k_j})_{\mathrm{n}}(x) \leq 0$, we obtain

$$v_{\mathrm{n}}(x) \leq 0 \qquad \text{for a.e. } x \in \Gamma_C.$$

Thus $v \in K$, so K is closed. ■

3.2.2 Small Strain Elasticity

We start by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega; \mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\overline{\Omega}$ by u is quantified.

Definition 3.1 (Small Strain Tensor). *The* `(small) strain tensor` ε *is defined by*

$$\varepsilon : H^1(\Omega; \mathbb{R}^d) \to L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \qquad u \mapsto \varepsilon(u) := \frac{1}{2}\left(\nabla u + \nabla u^\top\right).$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* $\mathcal{A}$, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field $\mathcal{A}$ is prescribed by the material and may vary in space, describing a *heterogeneous* body.

Definition 3.2 (Elasticity tensor field). *An* `elasticity tensor field` *is a fourth order tensor*

$$\mathcal{A} : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$$

such that the coefficients satisfy the symmetries

$$\mathcal{A}_{ijkl}(x) = \mathcal{A}_{jikl}(x) = \mathcal{A}_{ijlk}(x) = \mathcal{A}_{klij}(x), \qquad \forall\, 1 \le i, j, k, l \le d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_{\mathcal{A}} > 0$ *such that for a.e.* $x \in \Omega$

$$s : \mathcal{A}(x) : s \ge \alpha_{\mathcal{A}}\,(s : s) \qquad \forall\, s \in \mathrm{Sym}^2(\mathbb{R}^d).$$

Moreover, there exists $C_{\mathcal{A}} > 0$ *such that for a.e.* $x \in \Omega$

$$|\mathcal{A}_{ijkl}(x)| \le C_{\mathcal{A}} \qquad \forall\, 1 \le i, j, k, l \le d.$$

The *internal* and *surface forces* in the body $\overline{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 3.3 (Cauchy stress tensor). *Let* $\mathcal{A}$ *be an elasticity tensor field. We define the* `(linear) Cauchy stress tensor` *by*

$$\sigma : H^1(\Omega; \mathbb{R}^d) \to L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \qquad u \mapsto \mathcal{A} : \varepsilon(u).$$

On V, we define the symmetric bilinear form

$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v)\, \mathrm{d}x, \qquad u, v \in V \tag{3.4}$$

representing the *virtual work of internal forces* and the linear bounded functional

$$\ell(v) := \int_\Omega f \cdot v\, \mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v\, \mathrm{d}s, \qquad v \in V, \tag{3.5}$$

with *volume force density* $f \in L^2(\Omega; \mathbb{R}^d)$ on the body and *surface loads* $F \in L^2(\Gamma_N; \mathbb{R}^d)$ on the Neumann boundary, representing the *virtual work of external forces*. In the following, we want to prove standard properties of a and ℓ. For this, we will need the following inequalities. The first inequality is a result of *Korn's second inequality* ([1, Theorem 3.3] and [1, Section 3.5.2]) and the *Peetre–Tartar* Lemma ([1, Lemma 3.5]). Therefore only the proof of the second one is given.

Lemma 3.1. *Assuming* $|\Gamma_D| > 0$*, there exists a constant* $C_K > 0$ *such that*

$$C_K^{-1}\, \|v\|_{1,\Omega} \le \|\varepsilon(v)\|_{0,\Omega} \le \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

Proof. We prove the second inequality. For all $v \in V$ we have

$$|\varepsilon(v)(x)| = \left|\frac{1}{2}(\nabla v(x) + \nabla v^\top)\right| \le \frac{1}{2}(|\nabla v(x)| + |\nabla v(x)^\top|) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|_{0,\Omega}^2 = \int_\Omega |\varepsilon(v)(x)|^2 \, \mathrm{d}x \leq \int_\Omega |\nabla v(x)|^2 \, \mathrm{d}x = \|\nabla v\|_{0,\Omega}^2 \leq \|v\|_{1,\Omega}^2,$$

where the last inequality holds due to

$$\|v\|_{1,\Omega}^2 = \|v\|_{0,\Omega}^2 + \|\nabla v\|_{0,\Omega}^2.$$

■

Proposition 3.2. *The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined in (3.4) is*

i) bounded, *i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u,v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \qquad \text{with} \qquad C_1 = C_{\mathcal{A}},$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v,v) \geq C_2 \|v\|_{1,\Omega}^2 \qquad \text{with} \qquad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined in (3.5) is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \qquad \text{with} \qquad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \tag{3.6}$$

Proof. Let $v, w \in V$ be arbitrary. Then

$$|a(v,w)| \leq \|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega}$$

by the Cauchy–Schwarz inequality. With Definition 3.2 there exists $C_{\mathcal{A}} > 0$ such that

$$\|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}.$$

Combining this with the second inequality form Lemma 3.1 we have

$$|a(v,w)| \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega} \leq C_{\mathcal{A}} \|v\|_{1,\Omega} \|w\|_{1,\Omega}.$$

The ellipticity of a follows by

$$a(v,v) = \int_\Omega \varepsilon(v) : \mathcal{A} : \varepsilon(v) \, \mathrm{d}x \geq \alpha_{\mathcal{A}} \int_\Omega \varepsilon(v) : \varepsilon(v) = \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2,$$

where $\alpha_{\mathcal{A}} > 0$ exists due to Definition 3.2. The first inequality of Lemma 3.1 finally yields

$$a(v,v) \geq \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2 \geq \alpha_{\mathcal{A}} \left(C_K^{-1} \|v\|_{1,\Omega}\right)^2 = \frac{\alpha_{\mathcal{A}}}{C_K^2} \|v\|_{1,\Omega}^2.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{0,\Omega} \|v\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \|\gamma v\|_{0,\Gamma_N} \qquad \forall\, v \in V.$$

By the Trace Theorem 2.1 there exists $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma v\|_{0,\Gamma_N} \leq \|\gamma v\|_{0,\Gamma} \leq C_{\mathrm{tr}} \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

Finally,

$$|\ell(v)| \leq \left(\|f\|_{0,\Omega} + C_{\mathrm{tr}} \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega} \leq \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

■

3.2.3 Energy Minimization and Well-posedness

The total potential energy of the elastic body is given by the quadratic functional

$$\mathcal{J}(v) := \frac{1}{2}\, a(v,v) - \ell(v), \qquad v \in V. \tag{3.7}$$

In the presence of unilateral contact, the admissible displacements are restricted to the closed convex set $K \subset V$. We therefore consider the following constrained minimization problem.

Problem 3.1 (Signorini's Problem – Energy Minimization). *Find $u \in K$ such that*

$$\mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \frac{1}{2}\, a(v,v) - \ell(v).$$

We now what to prove well-posedness of the above Problem 3.1. This consist of existence and uniqueness of a solution and continuous dependency of the solution on the load functional ℓ. We start by the uniqueness, which follows directly by the standard arguments of strict convexity of $\mathcal{J}$ and the convexity of K. The latter was already proven in Proposition 3.1.

Lemma 3.2 (Strict Convexity of $\mathcal{J}$). *The Functional $\mathcal{J}$ defined in (3.7) is strictly convex.*

Proof. For $u, v \in V$ and $t \in [0,1]$, we set $w_t := (1-t)u + tv$. Then

$$a(w_t, w_t) = (1-t)a(u,u) + ta(v,v) - t(1-t)a(u-v, u-v)$$

and

$$\ell(w_t) = (1-t)\ell(u) + t\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1-t)\mathcal{J}(u) + t\mathcal{J}(v) - \frac{1}{2}t(1-t)a(u-v, u-v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V-ellipticity of a we have $a(u-v, u-v) > 0$ and $t(1-t) > 0$. This implies

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u) + t\mathcal{J}(v).$$

■

Proposition 3.3 (Uniqueness of Minimizer of $\mathcal{J}$). *If there exists a minimizer of the functional $\mathcal{J}$ defined in (3.7), then it is unique.*

Proof. Assume that there exist two minimizers $u_1, u_2 \in K$ of $\mathcal{J}$ on K, i.e.

$$\mathcal{J}(u_1) = \min_{v \in K} \mathcal{J}(v) = \mathcal{J}(u_2). \tag{3.8}$$

If $u_1 = u_2$ there is nothing to show. Hence assume $u_1 \neq u_2$. Since K is convex, for every $t \in (0,1)$ the convex combination

$$w_t := (1-t)u_1 + tu_2$$

belongs to K. By Lemma 3.2, $\mathcal{J}$ is strictly convex and therefore we have

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u_1) + t\mathcal{J}(u_2) \qquad \forall\, t \in (0,1).$$

Using (3.8), this yields

$$\mathcal{J}(w_t) < (1-t)\min_{v \in K} \mathcal{J}(v) + t \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \mathcal{J}(v).$$

This contradicts the definition of the minimum, since $w_t \in K$. Therefore the assumption $u_1 \neq u_2$ is false and we conclude $u_1 = u_2$. ■

It remains to show the existence of at least one minimizer. For this, we verify *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$. The proof requires the following lemma. Weak convergence is denoted by $\rightharpoonup$.

Lemma 3.3 (Weak lower semicontinuity of the norm)**.** *Let* $(X, \|\cdot\|_X)$ *be a normed space. Then* $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ *is a weakly lower semicontinuous functional, i.e. if* $x_j \rightharpoonup x$ *in* X*, then*

$$\|x\|_X \leq \liminf_{j\to\infty} \|x_j\|_X.$$

Proof. Let $J : X \to X^{**}$ be the canonical embedding defined by

$$(Jx)(\varphi) := \varphi(x) \qquad \forall\, \varphi \in X^*.$$

By the Hahn–Banach theorem, J is an isometry, i.e. $\|Jx\|_{X^{**}} = \|x\|_X$ for all $x \in X$. Therefore,

$$\|x\|_X = \|Jx\|_{X^{**}} = \sup_{\substack{\varphi\in X^*\\ \|\varphi\|_{X^*}\leq 1}} |(Jx)(\varphi)| = \sup_{\substack{\varphi\in X^*\\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x)|.$$

If $x_j \rightharpoonup x$ in X, then $\varphi(x_j) \to \varphi(x)$ for all $\varphi \in X^*$ and hence

$$\|x\|_X = \sup_{\substack{\varphi\in X^*\\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x)| \leq \liminf_{j\to\infty} \sup_{\substack{\varphi\in X^*\\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x_j)| = \liminf_{j\to\infty} \|x_j\|_X.$$

■

Lemma 3.4 ($\mathcal{J}$ is coercive and w.l.s.c.)**.** *The functional* $\mathcal{J}$ *defined in* (3.7) *is*

i) coercive *on* V*, i.e.*

$$\mathcal{J}(v) \to \infty \qquad \text{as} \qquad \|v\|_{1,\Omega} \to \infty,$$

ii) weakly lower semicontinuous *on* V*, i.e. if* $v_j \rightharpoonup v$ *in* V*, then*

$$\mathcal{J}(v) \leq \liminf_{j\to\infty} \mathcal{J}(v_j).$$

Proof. By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'}\|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \geq \frac{\alpha}{2}\, \|v\|_{1,\Omega}^2 - \|\ell\|_{V'}\, \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega}\to\infty} \infty.$$

For $s \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ set

$$\|s\|_{\mathcal{A}}^2 := \int_\Omega s : \mathcal{A} : s \,\mathrm{d}x.$$

By uniform ellipticity of $\mathcal{A}$, the map $\|\cdot\|_{\mathcal{A}}$ defines a norm on $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ equivalent to $\|\cdot\|_{0,\Omega}$. Moreover, $a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2$. The mapping $v \mapsto \varepsilon(v)$ is linear and bounded from V to $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$. Hence, if $v_j \rightharpoonup v$ in V, then

$$\varepsilon(v_j) \rightharpoonup \varepsilon(v) \qquad \text{in } L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Since norms are weakly lower semicontinuous by Lemma 3.3, we obtain

$$a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2 \leq \liminf_{j\to\infty} \|\varepsilon(v_j)\|_{\mathcal{A}}^2 = \liminf_{j\to\infty} a(v_j, v_j).$$

Furthermore, $\ell \in V'$ is weakly continuous, hence $\ell(v_j) \to \ell(v)$. Therefore,

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \leq \frac{1}{2} \liminf_{j\to\infty} a(v_j, v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

■

Proposition 3.4 (Existence of Minimizer of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in 3.7 has at least one minimizer $u \in K$.*

Proof. Set
$$M := \inf_{v \in K} \mathcal{J}(v) \in [-\infty, \infty).$$
Since $0 \in K$ we have $M \leq \mathcal{J}(0) = 0$, hence M is finite from above. Let $\{v_j\}_{j\in\mathbb{N}} \subset K$ be a minimizing sequence, i.e. $\mathcal{J}(v_j) \to M$ as $j \to \infty$. By Lemma 3.4 *i)*, $\mathcal{J}$ is coercive and thus, the sequence $\{v_j\}_{j\in\mathbb{N}}$ is bounded in V. Since V is a Hilbert space, there exist a subsequence $\{v_{j_k}\}_{k\in\mathbb{N}}$ and some $u \in V$ such that

$$v_{j_k} \rightharpoonup u \qquad \text{in } V.$$

As $K \subset V$ is closed and convex, it is weakly closed in V. Hence $u \in K$. By Lemma 3.4 *ii)*, $\mathcal{J}$ is weakly lower semicontinuity and thus we obtain

$$\mathcal{J}(u) \leq \liminf_{k\to\infty} \mathcal{J}(v_{j_k}) = M = \inf_{v\in K} \mathcal{J}(v),$$

so u is a minimizer of $\mathcal{J}$ on K. ∎

We collect the previous results in the following corollary and further give a standard stability estimation for the norm of the unique solution.

Corollary 3.1 (Well-posedness of Problem 3.1)**.** *The functional $\mathcal{J}$ defined in 3.7 admits a unique minimizer $u \in K$. Moreover, the minimizer satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2\,C_K^2\,\max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right). \tag{3.9}$$

Consequently, Problem 3.1 has a unique stable solution.

Proof. By Proposition 3.4 we have existence of a solution $u \in K$ to Problem 3.1 and by Proposition 3.3 it is unique. It remains to prove the a-priori bound. Since u is the unique minimizer, it holds
$$\mathcal{J}(u) \leq \mathcal{J}(0) = 0 \qquad \Longrightarrow \qquad \frac{1}{2}\,a(u,u) \leq \ell(u).$$
By Proposition 3.2 *ii)* we have
$$a(u,u) \geq \frac{\alpha_{\mathcal{A}}}{C_K^2}\,\|u\|_{1,\Omega}^2$$
and thus

$$\frac{\alpha_{\mathcal{A}}}{2\,C_K^2}\,\|u\|_{1,\Omega}^2 \leq \frac{1}{2}\,a(u,u) \leq |\ell(u)| \leq \max\{1, C_{\mathrm{tr}}\}\left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right)\|u\|_{1,\Omega},$$

where the last inequality is (3.6). Dividing both sides by $\|u\|_{1,\Omega}$ yields the desired estimate (3.9). For $u = 0$ the estimate is trivial. ∎

3.2.4 Variational inequality as an optimality condition

Finally, we turn to the *weak formulation* of Signorini's problem. Unlike the weak formulation of conventional simple boundary value problems, here we do not end up with a variational equation, but with a variational inequality. This is derived by a first-order optimality condition on the functional $\mathcal{J}$, i.e. a condition on its *Fréchet derivative*.

Definition 3.4 (Fréchet derivative). *Let X and Y be normed vector spaces and let*

$$f : U \subseteq X \to Y$$

be a function defined on an open subset U of X. Then f is called `Fréchet differentiable` *at $x \in U$ if there exists a bounded linear operator $A(x) : X \to Y$ such that*

$$\lim_{\|h\|_X \to 0} \frac{\|f(x+h) - f(x) - A(x)[h]\|_Y}{\|h\|_X} = 0. \tag{3.10}$$

If $A(x)$ exists, it is unique and is called the `Fréchet derivative` *of f at x. We write*

$$f'(x)[h] = A(x)[h] \qquad \forall\, h \in X.$$

Lemma 3.5 (Fréchet differentiability of $\mathcal{J}$). *The functional $\mathcal{J}$ defined in (3.7) is Fréchet differentiable on V. Its derivative at $u \in V$ is given by*

$$\mathcal{J}'(u)[h] = a(u,h) - \ell(h) \qquad \forall\, h \in V. \tag{3.11}$$

Proof. Let $u, k \in V$. Using bilinearity of a and linearity of ℓ, we obtain

$$\begin{aligned} \mathcal{J}(u+k) - \mathcal{J}(u) &= \frac{1}{2}\big(a(u+k,u+k) - a(u,u)\big) - \ell(k) \\ &= \frac{1}{2}\big(a(u,u) + 2a(u,k) + a(k,k) - a(u,u)\big) - \ell(k) \\ &= a(u,k) - \ell(k) + \frac{1}{2}\,a(k,k). \end{aligned}$$

Hence

$$\mathcal{J}(u+k) - \mathcal{J}(u) - \big(a(u,k) - \ell(k)\big) = \frac{1}{2}\,a(k,k).$$

Define $A(u) : V \to \mathbb{R}$ by

$$A(u)[h] := a(u,h) - \ell(h).$$

Since a and ℓ are bounded, $A(u)$ is linear and bounded. Moreover,

$$\frac{|\mathcal{J}(u+k) - \mathcal{J}(u) - A(u)[k]|}{\|k\|_{1,\Omega}} = \frac{1}{2}\,\frac{|a(k,k)|}{\|k\|_{1,\Omega}} \le \frac{1}{2}\,C_{\mathcal{A}}\,\|k\|_{1,\Omega} \xrightarrow{\|k\|_{1,\Omega}\to 0} 0,$$

by boundedness of a, cf. Proposition 3.2. Therefore, $\mathcal{J}$ is Fréchet differentiable at u with

$$\mathcal{J}'(u)[h] = A(u)[h] = a(u,h) - \ell(h) \qquad \forall\, h \in V.$$

■

Proposition 3.5 (First-order optimality condition on convex sets). *A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v-u] \ge 0 \qquad \forall\, v \in K. \tag{3.12}$$

Proof. Assume first that $u \in K$ minimizes $\mathcal{J}$ over K and let $v \in K$. By convexity of K we have $u + t(v-u) \in K$ for all $t \in [0,1]$. Define $\phi(t) := \mathcal{J}(u + t(v-u))$. Then ϕ attains its minimum at $t = 0$, hence $\phi'(0^+) \ge 0$, where $\phi'(0^+)$ denotes the right-sided derivative at $t = 0$. Using the chain rule and Fréchet differentiability of $\mathcal{J}$ we obtain

$$\phi'(0^+) = \mathcal{J}'(u)[v-u] \ge 0.$$

Conversely, assume (3.12). Let $v \in K$ be arbitrary and set $h := v - u$. By Taylor expansion of the quadratic functional $\mathcal{J}$ we have

$$\mathcal{J}(u+h) - \mathcal{J}(u) = \mathcal{J}'(u)[h] + \frac{1}{2} a(h,h).$$

By (3.12), $\mathcal{J}'(u)[h] \geq 0$, and since $a(h,h) \geq 0$ we conclude $\mathcal{J}(v) \geq \mathcal{J}(u)$ for all $v \in K$. Hence u is a minimizer over K. ∎

Combining Lemma 3.5 and Proposition 3.5, $u \in K$ solves Problem 3.1 if and only if

$$\mathcal{J}'(u)[v-u] = a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K.$$

This is the first weak formulation of Signorini's problem as a variational inequality.

Problem 3.2 (Signorini's Problem – Weak Formulation). *Find $u \in K$ such that*

$$a(u, v-u) \geq \ell(v-u) \qquad \forall\, v \in K.$$

Well-posedness of Problem 3.2 is already ensured, since it is equivalent to the minimization Problem 3.1 by Proposition 3.5 in the sense that each solution to the one of the problems is a solution to the other problem. The solution is unique due to Corollary 3.1 with the stability estimate (3.9). Another useful equivalent formulation is the following.

Problem 3.3 (Signorini's Problem – Weak Formulation). *Find $u \in K$ such that*

$$\begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \quad \forall\, v \in K. \end{cases}$$

Proposition 3.6 (Equivalence of Problem 3.2 and Problem 3.3). *Any solution $u \in K$ to Problem 3.2 is also a solution to Problem 3.3 and vice versa.*

Proof. Let $u \in K$ solve Problem 3.3. Then

$$a(u, v-u) = a(u,v) - a(u,u) = a(u,v) - \ell(u) \geq \ell(v) - \ell(u) = \ell(v-u).$$

Assuming that $u \in K$ solves Problem 3.2 and testing $v = 0$ yields

$$a(u, -u) \geq \ell(-u) \qquad \Longleftrightarrow \qquad -a(u,u) \geq -\ell(u),$$

which directly implies

$$a(u,u) \leq \ell(u).$$

However, testing with $v = 2u$ yields $a(u,u) \geq \ell(u)$ and thus $a(u,u) = \ell(u)$. Furthermore, Problem 3.2 reads

$$a(u, v-u) = a(u,v) - a(u,u) \geq \ell(v-u) = \ell(v) - \ell(u),$$

which directly implies

$$a(u,v) \geq a(u,u) + \ell(v) - \ell(u).$$

Plugging in $a(u,u) = \ell(u)$ yields $a(u,v) \geq \ell(v)$. ∎

We close this section by providing another stability result on the dependency of solutions to varying source terms as input data.

Proposition 3.7 (Lipschitz dependence on the data)**.** *For $i \in \{1, 2\}$, let $f_i \in L^2(\Omega; \mathbb{R}^d)$ and $F_i \in L^2(\Gamma_N; \mathbb{R}^d)$, and let $\ell_i \in V'$ be the corresponding load functional, i.e.*

$$\ell_i(v) := \int_\Omega f_i \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F_i \cdot \gamma v \,\mathrm{d}s, \qquad v \in V.$$

Moreover, let $u_i \in K$ be the unique solution of Problem 3.2 associated with ℓ_i, i.e.

$$a(u_i, v - u_i) \geq \ell_i(v - u_i) \qquad \forall\, v \in K.$$

Then the solution depends Lipschitz continuously on the load functional, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2}{\alpha_{\mathcal{A}}} \|\ell_1 - \ell_2\|_{V'}. \tag{3.13}$$

In particular, it depends Lipschitz continuously on the data f_i, F_i in the sense that

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2 \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \Big(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \Big). \tag{3.14}$$

Proof. For u_1, choose $v = u_2 \in K$ in Problem 3.2. Then

$$a(u_1, u_2 - u_1) \geq \ell_1(u_2 - u_1).$$

Similarly, choosing $v = u_1 \in K$ for u_2 yields

$$a(u_2, u_1 - u_2) \geq \ell_2(u_1 - u_2).$$

Adding both inequalities and using symmetry of a, we obtain

$$a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2).$$

By Proposition 3.2, the bilinear form a is V-elliptic with constant $\alpha_{\mathcal{A}}/C_K^2$. Hence

$$\frac{\alpha_{\mathcal{A}}}{C_K^2} \|u_1 - u_2\|_{1,\Omega}^2 \leq a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2) \leq \|\ell_1 - \ell_2\|_{V'} \|u_1 - u_2\|_{1,\Omega}.$$

If $u_1 = u_2$, there is nothing to prove. Otherwise, dividing by $\|u_1 - u_2\|_{1,\Omega}$ yields (3.13). üMoreover,

$$(\ell_1 - \ell_2)(v) = \int_\Omega (f_1 - f_2) \cdot v \,\mathrm{d}x + \int_{\Gamma_N} (F_1 - F_2) \cdot \gamma v \,\mathrm{d}s \qquad \forall\, v \in V.$$

Therefore, Proposition 3.2 applied to the functional $\ell_1 - \ell_2$ implies

$$\|\ell_1 - \ell_2\|_{V'} \leq \max\{1, C_{\mathrm{tr}}\} \Big(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \Big).$$

Combining this with (3.13), we obtain (3.14). ∎

3.3 Strong Formulation

From the previous weak formulations of Signorini's problem, we know there exists a uniquely determined $u \in K$ that solves the problem. To derive a *strong* PDE formulation, we have to discuss what was $\sigma(u)\mathrm{n}$ means and in what sense the contact conditions on Γ_C can be described as equalities and inequalities.

3.3.1 Stress Regularity and Boundary Tractions

In the weak formulation, the solution satisfies only $u \in K \subset H^1(\Omega; \mathbb{R}^d)$. Hence

$$\varepsilon(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)) \qquad \text{and} \qquad \sigma(u) = \mathcal{A} : \varepsilon(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

This is sufficient to define the bilinear form

$$a(u, v) = \int_\Omega \sigma(u) : \varepsilon(v) \, \mathrm{d}x,$$

but not yet enough to interpret the equilibrium equation $-\operatorname{div} \sigma(u) = f$ in $L^2(\Omega; \mathbb{R}^d)$ or to define boundary tractions in a classical pointwise sense. To formulate a strong equilibrium equation and the corresponding boundary forces, we therefore assume the additional regularity

$$\sigma(u) \in H(\operatorname{div}; \Omega).$$

By the general results from Section 2.7, the normal trace operator

$$\gamma_\mathrm{n} : H(\operatorname{div}; \Omega) \to H^{-1/2}(\Gamma; \mathbb{R}^d)$$

is well-defined. Accordingly, for $\tau \in H(\operatorname{div}; \Omega)$, its boundary traction is denoted by

$$\tau\, \mathrm{n} := \gamma_\mathrm{n}(\tau) \in H^{-1/2}(\Gamma; \mathbb{R}^d),$$

which agrees with the classical expression in the smooth case. Since every stress tensor $\tau \in H(\operatorname{div}; \Omega)$ is symmetric, we have

$$\tau : \nabla v = \tau : \varepsilon(v) \qquad \forall\, v \in H^1(\Omega; \mathbb{R}^d).$$

Therefore, the Green formula (2.4) takes the specific form

$$\langle \tau\, \mathrm{n}, \gamma v \rangle_\Gamma = \int_\Omega \tau : \varepsilon(v) \, \mathrm{d}x + \int_\Omega (\operatorname{div} \tau) \cdot v \, \mathrm{d}x \qquad \forall\, v \in H^1(\Omega; \mathbb{R}^d). \tag{3.15}$$

In particular, if $\sigma(u) \in H(\operatorname{div}; \Omega)$, then

$$\langle \sigma(u)\mathrm{n}, \gamma v \rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \, \mathrm{d}x + \int_\Omega \operatorname{div} \sigma(u) \cdot v \, \mathrm{d}x \qquad \forall\, v \in H^1(\Omega; \mathbb{R}^d). \tag{3.16}$$

This identity will be used in the derivation of the strong Signorini conditions.

3.3.2 A Hybrid Formulation

The weak formulation in Problem 3.2 encodes the equilibrium equation and the contact conditions only implicitly through the variational inequality. Under the additional stress regularity $\sigma(u) \in H(\operatorname{div}; \Omega)$, these conditions can be separated into a bulk equation in Ω and boundary relations involving the traction $\sigma(u)\mathrm{n}$. This leads to the following hybrid formulation, which may be viewed as an intermediate form between the weak variational inequality and the strong Signorini conditions in the later subsections.

Problem 3.4 (Signorini's Problem – Hybrid Formulation)**.** *For given* $f \in L^2(\Omega; \mathbb{R}^d)$ *and*

$F \in L^2(\Gamma_N;\mathbb{R}^d)$, *find* $u \in K$ *such that* $\sigma(u) \in H(\mathrm{div};\Omega)$ *and*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, & (3.17a) \\ \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \int_{\Gamma_N} F \cdot \gamma u \,\mathrm{d}s, & & (3.17b) \\ \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma v \,\mathrm{d}s & \forall\, v \in K. & (3.17c) \end{cases}$$

We close this section by showing, that the hybrid formulation above is equivalent to the previous weak formulations in Problem 3.2 in order to prove that it is indeed a valid formulation of the Signorini problem.

Proposition 3.8 (Equivalence of hybrid and weak formulation)**.** *Assume that* $\sigma(u) \in H(\mathrm{div};\Omega)$. *Then* $u \in K$ *solves Problem 3.2 if and only if it solves Problem 3.4.*

Proof. Assume first that $u \in K$ solves Problem 3.2, i.e.

$$a(u, v - u) \geq \ell(v - u) \qquad \forall\, v \in K. \tag{3.18}$$

Let $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$. Since $\gamma\varphi = 0$ on Γ, we have $u \pm t\varphi \in K$ for all sufficiently small $t > 0$. Testing (3.18) with $v = u \pm t\varphi$ yields

$$a(u, \varphi) = \ell(\varphi).$$

Because φ has compact support in Ω, the boundary term in $\ell(\varphi)$ vanishes, and thus

$$a(u, \varphi) = \int_\Omega f \cdot \varphi \,\mathrm{d}x.$$

Moreover, since $\sigma(u)$ is symmetric,

$$a(u, \varphi) = \int_\Omega \sigma(u) : \varepsilon(\varphi) \,\mathrm{d}x = \int_\Omega \sigma(u) : \nabla\varphi \,\mathrm{d}x.$$

By definition of the distributional divergence, this means

$$\langle -\operatorname{div}\sigma(u), \varphi\rangle = \int_\Omega \sigma(u) : \nabla\varphi \,\mathrm{d}x = \int_\Omega f \cdot \varphi \,\mathrm{d}x = \langle f, \varphi\rangle \qquad \forall\, \varphi \in \mathcal{D}(\Omega;\mathbb{R}^d).$$

Hence,

$$-\operatorname{div}\sigma(u) = f \qquad \text{in } \mathcal{D}'(\Omega;\mathbb{R}^d).$$

Since $\sigma(u) \in H(\mathrm{div};\Omega)$, we have $\operatorname{div}\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$, and therefore (3.17a) holds. Next, let $v \in K$ and set $w := v - u$. By (3.16) and (3.17a),

$$a(u, w) = \langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma - \int_\Omega \operatorname{div}\sigma(u) \cdot w \,\mathrm{d}x = \langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma + \int_\Omega f \cdot w \,\mathrm{d}x.$$

On the other hand, (3.18) gives

$$a(u, w) \geq \ell(w) = \int_\Omega f \cdot w \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s.$$

Subtracting the volume term yields

$$\langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s,$$

i.e.
$$\langle\sigma(u)\mathrm{n},\gamma v\rangle_\Gamma-\langle\sigma(u)\mathrm{n},\gamma u\rangle_\Gamma\ge\int_{\Gamma_N}F\cdot\gamma(v-u)\,\mathrm{d}s\qquad\forall\,v\in K.$$
It remains to prove (3.17b). Since $0\in K$ and $2u\in K$, testing (3.18) with $v=0$ and $v=2u$ gives
$$a(u,u)=\ell(u).$$
Applying (3.16) with $v=u$ and using (3.17a), we obtain
$$a(u,u)=\langle\sigma(u)\mathrm{n},\gamma u\rangle_\Gamma+\int_\Omega f\cdot u\,\mathrm{d}x.$$
Together with
$$\ell(u)=\int_\Omega f\cdot u\,\mathrm{d}x+\int_{\Gamma_N}F\cdot\gamma u\,\mathrm{d}s,$$
this yields (3.17b). Consequently, the previous inequality becomes
$$\langle\sigma(u)\mathrm{n},\gamma v\rangle_\Gamma\ge\int_{\Gamma_N}F\cdot\gamma v\,\mathrm{d}s\qquad\forall\,v\in K,$$
which is exactly (3.17c). Thus u solves Problem 3.4. Conversely, assume that $u\in K$ solves Problem 3.4. Let $v\in K$ and set $w:=v-u$. Using (3.16) and (3.17a), we get
$$a(u,w)=\langle\sigma(u)\mathrm{n},\gamma w\rangle_\Gamma+\int_\Omega f\cdot w\,\mathrm{d}x.$$
Moreover, subtracting (3.17b) from (3.17c) yields
$$\langle\sigma(u)\mathrm{n},\gamma w\rangle_\Gamma\ge\int_{\Gamma_N}F\cdot\gamma w\,\mathrm{d}s.$$
Hence
$$a(u,w)\ge\int_\Omega f\cdot w\,\mathrm{d}x+\int_{\Gamma_N}F\cdot\gamma w\,\mathrm{d}s=\ell(w)=\ell(v-u).$$
Since $w=v-u$, this is exactly
$$a(u,v-u)\ge\ell(v-u)\qquad\forall\,v\in K,$$
i.e. u solves Problem 3.2. ■

3.3.3 Normal/Tangential Splitting and Contact Trace Spaces

The hybrid formulation derived above still encodes the contact interaction through a single boundary traction functional acting on traces. In order to recover the Signorini contact conditions on Γ_C in a meaningful weak sense, we now split traces and tractions into normal and tangential components and introduce the corresponding trace spaces on the contact boundary. Recall that for $v\in V$, the trace $\gamma v|_{\Gamma_C}$ is well-defined. We define the contact trace space by
$$W_C:=\{\gamma v|_{\Gamma_C}\mid v\in V\}\subset H^{1/2}(\Gamma_C;\mathbb{R}^d).$$
For $w\in W_C$, we define its normal and tangential components by
$$w_\mathrm{n}:=w\cdot\mathrm{n},\qquad w_t:=w-(w\cdot\mathrm{n})\mathrm{n}.$$
Accordingly, we introduce the normal and tangential trace spaces
$$W_{C,\mathrm{n}}:=\{w\cdot\mathrm{n}\mid w\in W_C\},\qquad W_{C,t}:=\{w\in W_C\mid w\cdot\mathrm{n}=0\}.$$

Hence every $w \in W_C$ admits the decomposition

$$w = w_{\mathrm{n}}\mathrm{n} + w_t \in W_{C,\mathrm{n}}\,\mathrm{n} \oplus W_{C,t}$$

We endow W_C, $W_{C,\mathrm{n}}$ and $W_{C,t}$ with the quotient norms induced by V, namely

$$\|w\|_{W_C} := \inf\left\{\|v\|_{1,\Omega} \mid v \in V,\ \gamma v|_{\Gamma_C} = w\right\}, \qquad w \in W_C, \tag{3.19}$$

and

$$\|\eta\|_{W_{C,\mathrm{n}}} := \inf\left\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C}) \cdot \mathrm{n} = \eta\right\}, \qquad \eta \in W_{C,\mathrm{n}}, \tag{3.20}$$

and

$$\|\zeta\|_{W_{C,t}} := \inf\left\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C})_t = \zeta\right\}, \qquad \zeta \in W_{C,t}. \tag{3.21}$$

We denote by W_C', $W_{C,\mathrm{n}}'$ and $W_{C,t}'$ the corresponding dual spaces, equipped with the operator norms. Their duality pairings are denoted by $\langle\cdot,\cdot\rangle_{\Gamma_C}$. Whenever no confusion can arise, we use the same notation $\langle\cdot,\cdot\rangle_{\Gamma_C}$ for the pairings on $W_C' \times W_C$, $W_{C,\mathrm{n}}' \times W_{C,\mathrm{n}}$ and $W_{C,t}' \times W_{C,t}$. In the hybrid formulation, the boundary traction on Γ_C is understood as a bounded linear functional on W_C. Hence, for a contact traction $\tau \in W_C'$, one can define normal and tangential components

$$\tau_{\mathrm{n}} \in W_{C,\mathrm{n}}' \quad \text{and} \quad \tau_t \in W_{C,t}', \tag{3.22}$$

such that, for every $w \in W_C$ with decomposition $w = w_{\mathrm{n}}\mathrm{n} + w_t$,

$$\langle \tau, w\rangle_{\Gamma_C} = \langle \tau_{\mathrm{n}}, w_{\mathrm{n}}\rangle_{\Gamma_C} + \langle \tau_t, w_t\rangle_{\Gamma_C}. \tag{3.23}$$

To express the Signorini contact condition on the normal component in dual form, we introduce the cone of admissible normal traces

$$W_{C,\mathrm{n}}^- := \left\{\eta \in W_{C,\mathrm{n}} \mid \eta \le 0 \text{ a.e. on } \Gamma_C\right\}, \tag{3.24}$$

and its dual cone of weakly nonpositive normal contact forces

$$\Lambda_C := \left\{\lambda \in W_{C,\mathrm{n}}' \;\middle|\; \langle \lambda, \eta\rangle_{\Gamma_C} \ge 0 \quad \forall\, \eta \in W_{C,\mathrm{n}}^-\right\}. \tag{3.25}$$

With these notations, the boundary inequality in the hybrid formulation can be decomposed into normal and tangential contributions on Γ_C. This allows us to state the Signorini contact conditions as a KKT-type system in the dual spaces $W_{C,\mathrm{n}}'$ and $W_{C,t}'$.

3.3.4 KKT Contact Conditions

Under the assumption of having no Neumann boundary part, $|\Gamma_N|$, we can make a strong formulation of Signorini's Problem. We denote the boundary stress by

$$\sigma(u)\mathrm{n} = \sigma_{\mathrm{n}}(u)\mathrm{n} + \sigma_t(u).$$

Problem 3.5 (Signorini's Problem – Strong Formulation)**.** *For given $f \in L^2(\Omega;\mathbb{R}^d)$, find $u \in V$ such that*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, & (3.26a)\\ \sigma_t(u) = 0 & \text{in } W_{C,t}', & (3.26b)\\ u_{\mathrm{n}} \le 0 & \text{a.e. on } \Gamma_C, & (3.26c)\\ \sigma_{\mathrm{n}}(u) \in \Lambda_C, & & (3.26d)\\ \langle \sigma_{\mathrm{n}}(u),\, u_{\mathrm{n}}\rangle_{\Gamma_C} = 0. & & (3.26e)\end{cases}$$

Proposition 3.9 (Equivalence of Strong and Weak Formulation)**.** *Suppose that* $|\Gamma_N| = 0$. *Then every solution to Problem 3.5 is also a solution to Problem 3.4 and vice versa.*

Proof. We prove directly the equivalence between Problems 3.4 and 3.5. Assume first that $u \in V$ solves Problem 3.5. Since $u \in V$, we have

$$\sigma(u) = \mathcal{A} : \varepsilon(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Moreover, (3.26a) yields

$$\operatorname{div} \sigma(u) = -f \in L^2(\Omega; \mathbb{R}^d).$$

Hence $\sigma(u) \in H(\mathrm{div}; \Omega)$. In particular, (3.17a) holds. Now let $v \in K$. Since $|\Gamma_N| = 0$ and $\gamma v = 0$ on Γ_D, we have

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \langle \sigma(u)\mathrm{n}, \gamma v\rangle_{\Gamma_C}.$$

Using the decomposition into normal and tangential components, we get

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), v_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_t(u), v_t\rangle_{\Gamma_C}.$$

By (3.26b), the second term vanishes. Therefore,

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \langle \sigma_\mathrm{n}(u), v_\mathrm{n}\rangle_{\Gamma_C}.$$

Since $v \in K$, we have $v_\mathrm{n} \le 0$ a.e. on Γ_C, and by (3.26d) we know that $\sigma_\mathrm{n}(u) \in \Lambda_C$. Hence

$$\langle \sigma_\mathrm{n}(u), v_\mathrm{n}\rangle_{\Gamma_C} \ge 0,$$

which proves (3.17c), because $|\Gamma_N| = 0$. Taking $v = u$ and using (3.26e), we similarly obtain

$$\langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \langle \sigma_\mathrm{n}(u), u_\mathrm{n}\rangle_{\Gamma_C} = 0,$$

which is exactly (3.17b). Thus u solves Problem 3.4. Conversely, assume that $u \in K$ solves Problem 3.4. Then $u \in V$, and (3.26c) is already contained in the definition of K. Moreover, (3.17a) is exactly (3.26a). We next prove (3.26b). Let $w_t \in W_{C,t}$. By definition of $W_{C,t}$, there exists $w \in V$ such that

$$\gamma w|_{\Gamma_C} = w_t \qquad \text{and} \qquad w_\mathrm{n} = 0.$$

Since $u \in K$ and $w_\mathrm{n} = 0$, we have $u \pm w \in K$. Hence (3.17c) applied to $v = u + w$ and $v = u - w$, together with (3.17b), yields

$$\langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma \ge 0 \qquad \text{and} \qquad \langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma \le 0.$$

Therefore,

$$\langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma = 0.$$

Again using $|\Gamma_N| = 0$ and $\gamma w = 0$ on Γ_D, we obtain

$$0 = \langle \sigma(u)\mathrm{n}, \gamma w\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), w_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_t(u), w_t\rangle_{\Gamma_C} = \langle \sigma_t(u), w_t\rangle_{\Gamma_C}.$$

Since $w_t \in W_{C,t}$ was arbitrary, this proves (3.26b). We now prove (3.26d). Let $\eta \in W_C$ satisfy $\eta \le 0$ a.e. on Γ_C. By definition of W_C, there exists $v \in V$ such that $v_\mathrm{n} = \eta$. Because $\eta \le 0$, we have $v \in K$. Hence (3.17c) gives

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \ge 0.$$

Using $|\Gamma_N| = 0$, $\gamma v = 0$ on Γ_D, and (3.26b), we infer

$$0 \le \langle \sigma(u)\mathrm{n}, \gamma v\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), v_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_t(u), v_t\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), \eta\rangle_{\Gamma_C}.$$

This proves $\sigma_\mathrm{n}(u) \in \Lambda_C$, i.e. (3.26d). Finally, (3.17b), together with $|\Gamma_N| = 0$, $\gamma u = 0$ on Γ_D, and (3.26b), yields

$$0 = \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \langle \sigma(u)\mathrm{n}, \gamma u\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), u_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_t(u), u_t\rangle_{\Gamma_C} = \langle \sigma_\mathrm{n}(u), u_\mathrm{n}\rangle_{\Gamma_C},$$

which is exactly (3.26e). Hence u solves Problem 3.5. ∎

3.3.5 Classical Strong Form under Additional Regularity

Finally, under higher regularity assumptions on u, we can formulate Signorini's problem in a natural strong way. Here $|\Gamma_N| > 0$ is allowed.

Problem 3.6 (Signorini's Problem – Classical Strong Formulation). *For given $f \in L^2(\Omega;\mathbb{R}^d)$ and $F \in L^2(\Gamma_N;\mathbb{R}^d)$ find $u \in H^s(\Omega;\mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{in } \Omega, & (3.27a)\\ \sigma(u) = \mathcal{A} : \varepsilon(u) & \text{in } \Omega, & (3.27b)\\ u = 0 & \text{on } \Gamma_D, & (3.27c)\\ \sigma(u)\mathrm{n} = F & \text{on } \Gamma_N, & (3.27d)\end{cases}$$

and the contact conditions

$$\begin{cases} u_{\mathrm{n}} \le 0 & \text{on } \Gamma_C, & (3.28a)\\ \sigma_{\mathrm{n}}(u) \le 0 & \text{on } \Gamma_C, & (3.28b)\\ \sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} = 0 & \text{on } \Gamma_C, & (3.28c)\\ \sigma_t(u) = 0 & \text{on } \Gamma_C & (3.28d)\end{cases}$$

hold.

To prove equivalence to one of the previous formulations, we need the following Lemma.

Lemma 3.6 (Localized boundary liftings). *Assume the boundary decomposition is sufficiently regular. Then the following hold:*

(i) For every $g_N \in H^{1/2}(\Gamma_N;\mathbb{R}^d)$ there exists $w \in V$ such that

$$\gamma w = g_N \text{ on } \Gamma_N, \qquad \gamma w = 0 \text{ on } \Gamma_C.$$

(ii) For every $\zeta \in W_{C,t}$ there exists $w \in V$ such that

$$\gamma w = 0 \text{ on } \Gamma_N, \qquad w_{\mathrm{n}} = 0, \quad w_t = \zeta \text{ on } \Gamma_C.$$

(iii) For every $\eta \in W_{C,\mathrm{n}}$ there exists $w \in V$ such that

$$\gamma w = 0 \text{ on } \Gamma_N, \qquad w_t = 0, \quad w_{\mathrm{n}} = \eta \text{ on } \Gamma_C.$$

Proof. We repeatedly use the lifting operator on boundary parts provided by the trace and lifting theory from the mathematical framework. For (i), let $g_N \in H^{1/2}(\Gamma_N;\mathbb{R}^d)$. By the regularity assumption on the boundary decomposition, its zero extension

$$\operatorname{Ext}_0(g_N) := \begin{cases} g_N & \text{on } \Gamma_N,\\ 0 & \text{on } \Gamma \setminus \Gamma_N,\end{cases}$$

belongs to $H^{1/2}(\Gamma;\mathbb{R}^d)$. Let $w \in H^1(\Omega;\mathbb{R}^d)$ be a lifting of $\operatorname{Ext}_0(g_N)$, i.e. $\gamma w = \operatorname{Ext}_0(g_N)$ on Γ. Then, by construction,

$$\gamma w = g_N \text{ on } \Gamma_N, \qquad \gamma w = 0 \text{ on } \Gamma_C,$$

and also $\gamma w = 0$ on Γ_D. Hence $w \in V$. This proves (i). For (ii), let $\zeta \in W_{C,t}$. By definition of $W_{C,t}$, ζ is a tangential trace on Γ_C, hence

$$\zeta \in H^{1/2}_{00}(\Gamma_C;\mathbb{R}^d) \qquad \text{and} \qquad \zeta \cdot \mathrm{n} = 0 \text{ on } \Gamma_C.$$

Let $\mathrm{Ext}_0(\zeta)$ denote the zero extension of ζ from Γ_C to Γ, and let $w \in H^1(\Omega; \mathbb{R}^d)$ be a lifting such that $\gamma w = \mathrm{Ext}_0(\zeta)$ on Γ. Then $\gamma w = 0$ on Γ_N and on Γ_C we have $\gamma w = \zeta$. Since ζ is tangential, it follows that

$$w_\mathrm{n} = (\gamma w) \cdot \mathrm{n} = \zeta \cdot \mathrm{n} = 0 \qquad \text{on } \Gamma_C.$$

Moreover,

$$w_t = \gamma w - w_\mathrm{n} \mathrm{n} = \zeta \qquad \text{on } \Gamma_C.$$

Finally, $\gamma w = 0$ on Γ_D, so $w \in V$. This proves (ii). For (iii), let $\eta \in W_{C,\mathrm{n}}$. Then the vector-valued boundary function $g := \eta\, \mathrm{n}$ belongs to $W_C \subset H^{1/2}_{00}(\Gamma_C; \mathbb{R}^d)$. Let $\mathrm{Ext}_0(g)$ denote its zero extension from Γ_C to Γ, and let $w \in H^1(\Omega; \mathbb{R}^d)$ be a lifting such that $\gamma w = \mathrm{Ext}_0(g)$ on Γ. Then $\gamma 2 = 0$ on Γ_N and on Γ_C we have $\gamma w = \eta\, \mathrm{n}$. Hence

$$w_\mathrm{n} = (\gamma w) \cdot \mathrm{n} = (\eta\, \mathrm{n}) \cdot \mathrm{n} = \eta \qquad \text{on } \Gamma_C,$$

while

$$w_t = \gamma w - w_\mathrm{n} \mathrm{n} = \eta\, \mathrm{n} - \eta\, \mathrm{n} = 0 \qquad \text{on } \Gamma_C.$$

Again, $\gamma w = 0$ on Γ_D, so $w \in V$. This proves (iii). ∎

Proposition 3.10 (Equivalence of classical strong and hybrid formulation). *Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves Problem 3.6 if and only if it solves Problem 3.4.*

Proof. The additional regularity $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ is used precisely to interpret the boundary conditions in Problem 3.6 in the classical sense. Indeed, by the trace theorem,

$$\gamma u \in H^{s-1/2}(\Gamma; \mathbb{R}^d) \subset L^2(\Gamma; \mathbb{R}^d),$$

and therefore the normal and tangential components

$$u_\mathrm{n} := (\gamma u) \cdot \mathrm{n}, \qquad u_t := \gamma u - u_\mathrm{n} \mathrm{n}$$

are L^2-functions on Γ_C. In particular, the contact conditions in Problem 3.6 are genuine almost-everywhere conditions on Γ_C. Assume first that u solves Problem 3.6. Since $-\operatorname{div} \sigma(u) = f$ almost everywhere in Ω, the Green formula yields

$$\int_\Omega \sigma(u) : \varepsilon(v)\, \mathrm{d}x = \int_\Omega f \cdot v\, \mathrm{d}x + \langle \sigma(u)\mathrm{n}, \gamma v \rangle_\Gamma \qquad \forall\, v \in H^1(\Omega; \mathbb{R}^d).$$

Because in the present regularity regime the boundary traction is a classical boundary function, the duality pairing can be written as a boundary integral. Hence

$$a(u, v) = \int_\Omega f \cdot v\, \mathrm{d}x + \int_{\Gamma_N} \sigma(u)\mathrm{n} \cdot \gamma v\, \mathrm{d}s + \int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma v\, \mathrm{d}s.$$

Using the Neumann condition in Problem 3.6, this becomes

$$a(u, v) = \int_\Omega f \cdot v\, \mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v\, \mathrm{d}s + \int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma v\, \mathrm{d}s.$$

Let now $v \in K$. On Γ_C, we decompose

$$\sigma(u)\mathrm{n} \cdot \gamma v = \sigma_\mathrm{n}(u)\, v_\mathrm{n} + \sigma_t(u) \cdot v_t.$$

By the classical Signorini conditions (3.28a),(3.28b) and (3.28d),

$$\sigma_t(u) = 0, \qquad \sigma_\mathrm{n}(u) \le 0, \qquad v_\mathrm{n} \le 0 \quad \text{a.e. on } \Gamma_C,$$

and therefore

$$\int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma v \, \mathrm{d}s = \int_{\Gamma_C} \sigma_{\mathrm{n}}(u) \, v_{\mathrm{n}} \, \mathrm{d}s \geq 0.$$

Thus,

$$a(u,v) \geq \int_{\Omega} f \cdot v \, \mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v \, \mathrm{d}s = \ell(v) \qquad \forall\, v \in K.$$

This is exactly the hybrid inequality (3.17c). To prove the hybrid equality (3.17b), we take $v = u$. Then we have

$$\int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma u \, \mathrm{d}s = \int_{\Gamma_C} \sigma_{\mathrm{n}}(u) \, u_{\mathrm{n}} \, \mathrm{d}s + \int_{\Gamma_C} \sigma_t(u) \cdot u_t \, \mathrm{d}s.$$

Again by the classical Signorini conditions (3.28d) and (3.28c),

$$\sigma_t(u) = 0 \qquad \text{and} \qquad \sigma_{\mathrm{n}}(u) \, u_{\mathrm{n}} = 0 \quad \text{a.e. on } \Gamma_C,$$

and thus

$$\int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma u \, \mathrm{d}s = 0.$$

Hence

$$a(u,u) = \int_{\Omega} f \cdot u \, \mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma u \, \mathrm{d}s = \ell(u),$$

which is the hybrid equality (3.17b). Therefore u solves Problem 3.4. Conversely, assume that u solves Problem 3.4. The bulk equation

$$-\operatorname{div} \sigma(u) = f \qquad \text{a.e. in } \Omega$$

is already contained in the hybrid formulation. Moreover, since $u \in K \subset V$, we also obtain

$$u = 0 \text{ on } \Gamma_D \qquad \text{and} \qquad u_{\mathrm{n}} \leq 0 \text{ on } \Gamma_C,$$

which is exactly (3.27c) and (3.28a). It remains to prove the Neumann condition on Γ_N and the remaining contact conditions on Γ_C. We first prove the Neumann condition. Let $g_N \in H^{1/2}(\Gamma_N; \mathbb{R}^d)$ and choose $w \in V$ as in Lemma 3.6(i), i.e.

$$\gamma w = g_N \text{ on } \Gamma_N, \qquad \gamma w = 0 \text{ on } \Gamma_C.$$

Since $w_{\mathrm{n}} = 0$ on Γ_C, both w and $-w$ belong to K. Applying the hybrid inequality to w and to $-w$, we get

$$\langle \sigma(u)\mathrm{n}, \gamma w \rangle_{\Gamma} = \int_{\Gamma_N} F \cdot \gamma w \, \mathrm{d}s.$$

Because $\gamma w = 0$ on Γ_C, this reads

$$\int_{\Gamma_N} \sigma(u)\mathrm{n} \cdot g_N \, \mathrm{d}s = \int_{\Gamma_N} F \cdot g_N \, \mathrm{d}s \qquad \forall\, g_N \in H^{1/2}(\Gamma_N; \mathbb{R}^d).$$

Since both $\sigma(u)\mathrm{n}$ and F are L^2-functions on Γ_N, it follows that

$$\sigma(u)\mathrm{n} = F \qquad \text{a.e. on } \Gamma_N.$$

This proves the Neumann boundary condition (3.27d). Next, we prove the tangential contact condition (3.28d). Let $\zeta \in W_{C,t}$ and choose $w \in V$ according to Lemma 3.6(ii), i.e.

$$\gamma w = 0 \text{ on } \Gamma_N, \qquad w_{\mathrm{n}} = 0, \quad w_t = \zeta \text{ on } \Gamma_C.$$

Again, both w and $-w$ belong to K, and therefore the hybrid inequality applied to $\pm w$ yields

$$\langle \sigma(u)\mathrm{n}, \gamma w \rangle_{\Gamma} = 0.$$

Because $\gamma w = 0$ on Γ_N and $w_{\mathrm{n}} = 0$ on Γ_C, we obtain

$$0 = \int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot w_t \, \mathrm{d}s = \int_{\Gamma_C} \sigma_t(u) \cdot \zeta \, \mathrm{d}s \qquad \forall \zeta \in W_{C,t}.$$

Hence $\sigma_t(u) = 0$ almost everywhere on Γ_C which is (3.28d). We now prove the normal sign condition (3.28b). Let $\eta \in W_{C,\mathrm{n}}$ satisfy $\eta \leq 0$ almost everywhere on Γ_C, and choose $w \in V$ as in Lemma 3.6(iii), i.e.

$$\gamma w = 0 \text{ on } \Gamma_N, \qquad w_t = 0, \quad w_{\mathrm{n}} = \eta \text{ on } \Gamma_C.$$

Then $w \in K$, and the hybrid inequality gives

$$0 \leq \langle \sigma(u)\mathrm{n}, \gamma w \rangle_\Gamma = \int_{\Gamma_C} \sigma_{\mathrm{n}}(u)\, \eta \, \mathrm{d}s \qquad \forall \eta \in W_{C,\mathrm{n}}, \ \eta \leq 0.$$

Therefore, $\sigma_{\mathrm{n}}(u) \leq 0$ almost everywhere on Γ_C which is exactly (3.28b). Finally, we prove the complementarity condition (3.28c). The hybrid equality (3.17b) gives

$$\langle \sigma(u)\mathrm{n}, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot \gamma u \, \mathrm{d}s.$$

Using the Neumann condition (3.27d) already proved on Γ_N, we obtain

$$\int_{\Gamma_C} \sigma(u)\mathrm{n} \cdot \gamma u \, \mathrm{d}s = 0.$$

Since $\sigma_t(u) = 0$ on Γ_C by (3.28d), this reduces to

$$\int_{\Gamma_C} \sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} \, \mathrm{d}s = 0.$$

Now by (3.28a) and (3.28b) it is

$$u_{\mathrm{n}} \leq 0 \qquad \text{and} \qquad \sigma_{\mathrm{n}}(u) \leq 0 \qquad \text{a.e. on } \Gamma_C,$$

so the product $\sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}}$ is nonnegative almost everywhere on Γ_C. Since its integral is zero, it follows that

$$\sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} = 0 \qquad \text{a.e. on } \Gamma_C.$$

Thus all conditions in Problem 3.6 hold, and u solves the classical strong formulation. ■

3.4 Mixed Formulation

In the primal setting, Signorini's problem is posed as a variational inequality on the convex set $K \subset V$. In the mixed approach we introduce an additional unknown λ on the contact boundary, which represents the weak normal contact traction. This leads to a saddlepoint structure in the variables (u, λ), where the nonpenetration constraint is encoded by the cone condition $\lambda \in \Lambda_C$.

3.4.1 Mixed Weak Formulation as a Saddlepoint Problem

We keep the energy functional

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v), \qquad v \in V$$

from (3.7) and use the normal trace space $W_{C,\mathrm{n}}$ and the dual cone $\Lambda_C \subset W'_{C,\mathrm{n}}$ introduced in (3.25). For $v \in V$ and $\lambda \in W'_{C,\mathrm{n}}$ we define the coupling form

$$b(v, \lambda) := \langle \lambda, \, (\gamma v|_{\Gamma_C}) \cdot \mathrm{n} \rangle_{\Gamma_C}. \tag{3.29}$$

We then introduce the Lagrangian on $V \times \Lambda_C$ by

$$\mathcal{L}(v, \lambda) := \mathcal{J}(v) - b(v, \lambda). \tag{3.30}$$

Problem 3.7 (Signorini's Problem – Mixed weak formulation)**.** *Find* $(u, \lambda) \in V \times \Lambda_C$ *such that*

$$\begin{cases} a(u, v) - b(v, \lambda) = \ell(v) & \forall\, v \in V, \\ b(u, \mu - \lambda) \geq 0 & \forall\, \mu \in \Lambda_C. \end{cases}$$

The first condition (**??**) corresponds to bulk equilibrium with an additional boundary reaction term encoded by λ. The second condition (**??**) is a variational inequality on the cone Λ_C and represents the complementarity structure of frictionless contact.

Remark 3.1 (Saddlepoint property)**.** *A pair* $(u, \lambda) \in V \times \Lambda_C$ *solves Problem 3.7 if and only if it satisfies the saddlepoint inequality*

$$\mathcal{L}(u, \mu) \leq \mathcal{L}(u, \lambda) \leq \mathcal{L}(v, \lambda) \quad \forall\, (v, \mu) \in V \times \Lambda_C. \tag{3.31}$$

3.4.2 Well–posedness and inf–sup condition

We briefly explain the role of the inf–sup condition for the coupling form $b(\cdot, \cdot)$ in (3.29). Define the operators $A : V \to V'$ and $B : V \to W_{C,\mathrm{n}}$ by

$$(Au)(v) := a(u, v), \quad Bv := (\gamma v|_{\Gamma_C}) \cdot \mathrm{n}. \tag{3.32}$$

Then (**??**) can be written as

$$Au - B^*\lambda = \ell \quad \text{in } V', \tag{3.33}$$

where $B^* : W'_{C,\mathrm{n}} \to V'$ is the adjoint operator given by

$$(B^*\lambda)(v) = b(v, \lambda) \quad \forall\, v \in V. \tag{3.34}$$

In the purely linear saddlepoint setting (without the cone constraint $\lambda \in \Lambda_C$), well–posedness of the block system is governed by the inf–sup condition

$$\inf_{\lambda \in W'_{C,\mathrm{n}} \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{b(v, \lambda)}{\|v\|_{1,\Omega} \, \|\lambda\|_{W'_{C,\mathrm{n}}}} \geq \beta > 0. \tag{3.35}$$

Equivalently, (3.35) can be written as

$$\|B^*\lambda\|_{V'} \geq \beta \, \|\lambda\|_{W'_{C,\mathrm{n}}} \quad \forall\, \lambda \in W'_{C,\mathrm{n}}. \tag{3.36}$$

This expresses stability of the coupling between u and λ and, in particular, implies uniqueness of the multiplier once u is fixed.

Proposition 3.11 (Uniqueness of the multiplier)**.** *Assume that* (3.35) *holds. If* (u, λ_1) *and* (u, λ_2) *satisfy* (**??**) *with the same* $u \in V$*, then* $\lambda_1 = \lambda_2$ *in* $W'_{C,\mathrm{n}}$.

Proof. Subtracting (**??**) for λ_1 and λ_2 yields

$$b(v, \lambda_1 - \lambda_2) = 0 \quad \forall\, v \in V. \tag{3.37}$$

Hence $(B^*(\lambda_1 - \lambda_2))(v) = 0$ for all $v \in V$, i.e. $B^*(\lambda_1 - \lambda_2) = 0$ in V'. Using (3.36) gives $\lambda_1 - \lambda_2 = 0$. ∎

Finally, note that well–posedness of the mixed Signorini formulation also follows from its equivalence to the primal (or hybrid) formulation established earlier. Nevertheless, the inf–sup viewpoint is useful since it explains the saddlepoint structure and the stability of the multiplier.

4 Outlook

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

B List of Figures